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SURVEY

Privacy-Preserving Federated Learning for Intrusion Detection in IoT Environments: A Survey

ABHISHEK VYAS[1], (Graduate Student Member, IEEE), PO-CHING LIN[1], (Member, IEEE), REN-HUNG HWANG[2], (Senior Member, IEEE), AND MEENAKSHI TRIPATHI[3], (Senior Member, IEEE)

[1]Computer Science and Information Engineering Department, National Chung Cheng University, Minxiong, Chiayi 62102, Taiwan
[2]College of Artificial Intelligence, National Yang Ming Chiao Tung University, Tainan 30010, Taiwan
[3]Computer Science and Engineering Department, Malaviya National Institute of Technology Jaipur, Jaipur, Rajasthan 302017, India

Corresponding author: Abhishek Vyas (d09410004@ccu.edu.tw; abhishek_vyas_2022_7@ieee.org)

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ABSTRACT With the rapid development of artificial intelligence and a new generation of network technologies, the Internet of Things (IoT) is expanding worldwide. Malicious agents consistently exploit new technical vulnerabilities to access the various IoT systems used in critical industries, medical diagnosis, military, and defense systems. To mitigate these threats, IoT networks should be equipped with intrusion detection systems capable of detecting threat vectors in an attempt to compromise the systems. Moreover, many researchers have integrated privacy-preserving technologies such as homomorphic encryption, differential privacy, and secure multiparty computation with machine learning algorithms. Furthermore, federated learning, which shares only model parameters rather than data, provides distributed privacy-preserving learning; therefore, federated learning is secure and reliable for the implementation of intrusion detection systems in IoT environments. This survey examined the utilization and applications of privacy-preserving mechanisms, explicitly focusing on privacy-preserving federated learning for intrusion detection systems in IoT environments. This survey also highlights future research directions and open research questions. Privacy-preserving federated learning can significantly contribute to the rapid and efficient detection and prevention of various threat vectors that target IoT ecosystems.

INDEX TERMS Privacy preservation, federated learning, intrusion detection system, Internet of Things.

I. INTRODUCTION

It is a world of hyper-connectivity and automation, and many household and industrial devices are connected to the Internet. Machine learning applications that use data generated from various Internet-of-Things (IoT) devices in different domains, such as smart homes, smart vehicles, smart industries, and smart healthcare, show great promise. The current industrial revolution, Industry 5.0, combines AI and next-generation network technologies to produce smart, reliable, and scalable technologies, a significant step forward for human civilization.

Threats exist to the security and privacy of networks and data shared between various devices. This trend has prompted the development of multiple technologies to preserve the safety and confidentiality of data communication between different connected devices. The most prominent are cryptographic techniques, such as encryption, hashing, and digital signature. Software and hardware firewalls and intrusion detection/prevention systems (IDS/IPS) are used to prevent threat vectors from acting on specific networks. The survey covers and analyzes studies related to the combination of machine learning (ML), deep learning (DL), federated learning (FL), privacy-preserving technologies, and intrusion detection systems (IDS) to construct federated privacy-preserving learning for IDS in IoT. After surveying the available literature related to this paper's topic, we found

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that despite the various articles on the subsets of the subjects covered in the present survey, they have yet to cover a complete set. Table 1 lists the abbreviations and their full forms for some of the standard terms used in this survey.

In recent decades, there has been an exponential increase in the number of IoT devices in various forms. According to Finances Online, there will be approximately 25.44 billion connected online devices worldwide by 2030 [1]. Similarly, IoT analytics predicts that the number of devices installed online worldwide will reach 29.70 billion by 2029 [2]. The sheer number and variety of IoT devices and the growth in their installation will increase the surface and area of threat of the vulnerabilities to which IoT devices will be exposed. The number of attack scenarios alone will be enormous. Therefore, there is a need to explore ways to prevent the intrusion of malware, worms, and viruses into IoT devices at multiple levels. According to the literature surveyed in this paper, using privacy-preserving federated learning (PPFL) to implement IDS in IoT environments can be one of the most effective methods to eliminate intrusions in IoT environments.

A. ABBREVIATIONS AND ACRONYMS

TABLE 1. Abbreviations and full names.

Abbr.	Full Name
AI	Artificial Intelligence
AIDS	Anomaly-Based Intrusion Detection Systems
BIDS	Behavior-Based Intrusion Detection Systems
FL	Federated Learning
FTL	Federated Transfer Learning
ML	Machine Learning
DL	Deep Learning
RL	Reinforcement Learning
DP	Differential Privacy
GDPR	General Data Protection Regulation
HE	Homomorphic Encryption
HFL	Horizontal Federated Learning
HIDS	Host-Based Intrusion Detection Systems
HIPAA	Health Insurance Portability and Accountability Act
NIDS	Network-Based Intrusion Detection Systems
VFL	Vertical Federated Learning
PPL	Privacy-Preserving Learning
PPFL	Privacy-Preserving Federated Learning
PPML	Privacy-Preserving Machine Learning
PPDL	Privacy-Preserving Deep Learning
PPM	Privacy-Preserving Mechanism
IDS	Intrusion Detection Systems
NIDS	Network Intrusion Detection System
HIDS	Host-Based Intrusion Detection System
IID	Independent and Identically Distributed
Non-IID	Non-Independent and Identically Distributed
IDS	Intrusion Detection Systems
IPS	Intrusion Prevention Systems
IoT	Internet of Things
PPFL-IDS	PPFL for IDS
PPFL-IoT	PPFL for IoT
IIoT	Industrial Internet of Things
IoMT	Internet of Medical Things
SMPC	Secure Multi-Party Computation
TEE	Trusted Execution Environments
PPFL-IDS-IoT	PPFL for implementing IDS in IoT

B. SURVEY METHODOLOGY

For this study, the authors conducted a systematic search and review of the relevant research literature to identify relevant studies or literature on the composition of this survey. In trying to determine the relevant literature for the composition of this survey, the authors attempt to answer the following questions.

1) How many reviews and surveys have already been published on the topic related to this survey?
2) How can the available survey papers and literature on the topic be categorized to give readers a clear view of the survey?
3) What search strategies will be used to obtain relevant literature to formulate the survey?

The literature selection criteria of this survey included peer-reviewed conference papers, journal articles, and highly cited magazine articles relevant to the study. Consequently, the authors ensured the inclusion of pertinent high-quality literature. The databases include IEEE Xplore, ACM Digital Library, Scopus, and Google Scholar. The search strategy involved keyword searches within these databases, employing terms such as ''Federated Learning'' combined with ''Privacy-Preservation'' and ''Intrusion Detection,'' as well as ''Internet of Things.'' Duplicate search results were removed and the search scope was restricted to documents published in conferences, journals, and magazines. Non-English papers were excluded from the analysis. The authors conducted a preliminary review of all the abstracts of the retrieved literature to filter out materials that did not align with the research objectives of the survey. Consequently, this survey and review excluded papers lacking relevant and substantive content on the study of Federated Learning for the Protection of Privacy and its importance in intrusion detection systems for IoT environments.

C. SURVEY OBJECTIVES MOTIVATION AND JUSTIFICATION

This section delineates the objectives, needs, and rationales of this survey. Initially, the literature pertinent to the survey topic was organized by the issues addressed. Subsequent selections of survey papers and reviews from the cited literature substantiate the need for this specific survey. Table 2 classifies the research articles according to their focus areas: FL, PPFL, FL-IDS, PPFL-IDS, FL-IoT, PPFL-IoT, and PPFL-IDS-IoT, thus providing the readers with an overview of the topics discussed in this survey. Table 3 contrasts various published surveys, review articles, tutorials, and doctoral dissertations with the present survey, offering a comprehensive and analytical summary of the cutting-edge literature in PPFL. The comparison provides the readers with detailed information and underscores the significance of this survey. Both tables demonstrate that, while other research articles may address only selected topics, this survey comprehensively encompasses all pertinent issues, with a special focus on the application of PPFL in IDS for IoT environments. The tables effectively clarify the necessity of this survey, providing logical and categorical justifications.

TABLE 2. Topic categorization of the referred research articles.

Ref.	Ref. Type	FL	PPFL	FL-IDS	PPFL-IDS	FL-IoT	PPFL-IoT	PPFL-IDS-IoT
[11]	Magazine Article	✓	×	✓	×	✓	×	×
[19]	Journal Paper	✓	×	×	×	✓	×	×
[24]	Magazine Article	✓	✓	×	×	×	×	×
[31]	Journal Paper	✓	✓	×	×	✓	✓	×
[35]	Journal Paper	✓	✓	✓	×	×	×	×
[37]	Journal Paper	✓	×	×	×	×	×	×
[39]	Journal Paper	✓	✓	✓	✓	×	×	×
[40]	Journal Paper	✓	✓	×	×	✓	✓	×
[42]	Conference Paper	✓	✓	×	×	×	×	×
[44]	Journal Paper	✓	✓	×	×	✓	✓	×
[47]	Journal Paper	✓	×	✓	×	✓	×	×
[48]	Journal Paper	✓	✓	×	×	✓	✓	×
[49]	Journal Paper	✓	×	×	×	✓	×	×
[50]	Journal Paper	✓	×	✓	×	×	×	×
[51]	Journal Paper	✓	×	✓	×	✓	×	×
[52]	Journal Paper	✓	✓	×	×	×	✓	×
[53]	Conference Paper	✓	✓	×	×	×	×	×
[56]	Conference Paper	✓	✓	×	×	×	×	×
[58]	Journal Paper	✓	✓	×	×	×	×	×
[61]	Journal Paper	✓	✓	✓	×	✓	×	×
[63]	Journal Paper	✓	✓	×	×	×	×	×
[64]	Magazine Article	✓	✓	×	×	✓	✓	×
[65]	Journal Paper	✓	✓	×	×	×	×	×
[67]	Conference Paper	✓	✓	×	×	×	×	×
[68]	Journal Paper	✓	×	✓	×	✓	×	×
[69]	Journal Paper	✓	✓	×	×	×	×	×
[70]	Journal Paper	✓	✓	✓	✓	×	×	×
[72]	Conference Paper	✓	✓	×	×	×	×	×
[73]	Journal Paper	✓	✓	×	×	×	×	×
[74]	Journal Paper	✓	✓	✓	✓	×	×	×
[75]	Journal Paper	✓	✓	×	×	×	×	×
[76]	Conference Paper	✓	✓	✓	✓	×	×	×
[77]	Journal Paper	✓	✓	×	×	✓	✓	×
[78]	Journal Paper	✓	×	✓	×	✓	×	×
[79]	Journal Paper	✓	✓	✓	✓	×	×	×
[80]	Journal Paper	✓	×	✓	×	✓	×	×
[81]	Conference Paper	✓	✓	×	×	×	×	×
[82]	Journal Paper	✓	✓	✓	×	✓	×	×
[83]	Journal Paper	✓	×	✓	×	×	×	×
[84]	Journal Paper	✓	×	✓	×	×	×	×
[85]	Journal Paper	✓	✓	×	×	×	×	×
[86]	Conference Paper	✓	×	✓	×	✓	×	×
[87]	Journal Paper	✓	✓	×	×	✓	✓	×
[90]	Journal Paper	✓	✓	×	✓	×	✓	×
[91]	Conference Paper	✓	✓	×	×	✓	✓	×
[92]	Journal Paper	✓	✓	×	×	×	×	×
[94]	Conference Paper	✓	×	×	×	×	×	×
[96]	Journal Paper	✓	✓	✓	✓	×	×	×
[97]	Journal Paper	✓	✓	×	×	×	×	×
[98]	Journal Paper	✓	✓	✓	✓	×	×	×
[99]	Journal Paper	✓	✓	×	×	✓	✓	×
[100]	Journal Paper	✓	✓	✓	✓	×	✓	×
[103]	Conference Paper	✓	×	✓	×	✓	×	×
[107]	Journal Paper	✓	✓	×	×	✓	✓	×
[108]	Journal Paper	✓	✓	×	×	✓	✓	×
[112]	Journal Paper	✓	✓	×	×	×	×	×
[113]	Conference Paper	✓	✓	×	×	×	×	×
[114]	Journal Paper	✓	✓	×	×	×	×	×
[115]	Journal Paper	✓	✓	×	✓	×	✓	×
[116]	Journal Paper	✓	✓	×	×	×	✓	×
[117]	Journal Paper	✓	✓	×	×	×	✓	×
[118]	Journal Paper	✓	✓	×	×	×	×	×
[120]	Magazine Paper	✓	✓	×	×	×	✓	×
[121]	Journal Paper	✓	✓	×	×	×	×	×
[123]	Journal Paper	✓	✓	×	×	×	×	×
[124]	Conference Paper	✓	✓	×	×	×	×	×
[125]	Conference Paper	✓	✓	×	✓	×	✓	×
[126]	Journal Paper	✓	✓	×	×	✓	✓	×
[127]	Conference Paper	✓	✓	×	×	×	×	×
[128]	Journal Paper	✓	✓	×	×	×	×	×
[130]	Conference Paper	✓	✓	×	×	✓	✓	×
[133]	Journal Paper	✓	✓	×	×	×	×	×
[134]	Journal Paper	✓	✓	×	×	×	×	×
[135]	Journal Paper	✓	✓	×	×	×	×	×
[136]	Journal Paper	✓	✓	×	×	×	✓	×
[137]	Journal Paper	✓	✓	×	✓	×	×	×
[138]	Journal Paper	✓	×	✓	×	×	×	×
[139]	Journal Paper	✓	✓	×	×	✓	×	×
[140]	Journal Paper	✓	✓	×	×	×	✓	×
[141]	Conference Paper	✓	✓	×	×	×	×	×
[142]	Journal Paper	✓	✓	×	×	×	×	×
[143]	Journal Paper	✓	✓	×	×	×	×	×
[144]	Journal Paper	✓	✓	×	×	✓	✓	×
[145]	Conference Paper	✓	✓	×	×	×	×	×
[146]	Journal Paper	✓	✓	×	×	×	×	×
[147]	Journal Paper	✓	✓	×	×	×	×	×
[149]	Conference Paper	✓	×	✓	×	×	×	×
[151]	Journal Paper	✓	✓	×	×	×	×	×
[152]	Journal Paper	✓	×	✓	×	×	×	×
[154]	Journal Paper	✓	✓	×	×	✓	✓	×
[155]	Journal Paper	✓	×	×	×	×	×	×
[156]	Conference Paper	✓	✓	×	×	×	×	×
[157]	Journal Paper	✓	✓	×	×	×	×	×
[158]	Journal Paper	✓	✓	×	×	✓	✓	×
[159]	Journal Paper	✓	✓	✓	✓	×	×	×
[160]	Conference Paper	✓	✓	×	×	×	×	×
[161]	Journal Paper	✓	✓	×	×	✓	✓	×
[162]	Journal Paper	✓	✓	✓	✓	×	×	×
[163]	Journal Paper	✓	✓	×	×	×	×	×
[164]	Journal Paper	✓	✓	×	×	×	×	×
[166]	Journal Paper	✓	✓	×	×	×	×	×
[167]	Journal Paper	✓	×	✓	×	✓	×	×
[168]	Journal Paper	✓	✓	×	×	✓	✓	×
[169]	Conference Paper	✓	✓	×	×	×	×	×
[170]	Journal Paper	✓	✓	×	✓	×	✓	×
[172]	Journal Paper	✓	✓	×	✓	×	✓	×
[173]	Journal Paper	✓	×	×	×	×	×	×
[174]	Journal Paper	✓	✓	×	✓	×	✓	×
[178]	Journal Paper	✓	✓	×	×	✓	✓	×
[179]	Journal Paper	✓	✓	×	×	✓	✓	×
[182]	Journal Paper	✓	×	×	×	×	×	×
[183]	Journal Paper	✓	✓	×	×	×	×	×
This paper	Survey Paper	✓	✓	✓	✓	✓	✓	✓

The main objectives of this survey paper are listed as follows:

- This survey presents a state-of-the-art study of PPFL for IDS implementation in IoT systems.
- This survey justifies the need for research on PPFL for the implementation of IDS in IoT systems because this topic offers enhanced privacy preservation by minimizing exposure to sensitive user data while ensuring compliance with data protection regulations and improving efficiency by distributing the training process across multiple edge devices.
- In this survey, open research questions and future research directions are also discussed.

D. SURVEY STRUCTURE

Figure 1 illustrates the structure and organization of the survey. Section I serves as the introduction, where we familiarize the readers with the article and provide a comprehensive discussion of the necessity and motivation of this survey in the current context, based on research on the survey topic and the literature on related issues cited here. Section I elucidates the abbreviations and acronyms most frequently used throughout this paper. It outlines the systematic approach used by the authors to conduct a literature survey and compose this study. Section II introduces the reader to different privacy-preserving mechanisms that FL can use to enhance and supplement privacy to implement IDS in IoT systems. Section III is about FL, its various types and applications, and how PPFL is applied in multiple scenarios. This section elaborates on the numerous attack vectors that affect FL and also gives the readers a perspective on various FL aggregation algorithms and frameworks. Section IV discusses IDS in the context of IoT systems, various types of IDS, and their applications in multiple types of IoT networks. In addition, it covers the numerous types of IDS that are accessible for different types of system, including the Internet of Things platforms. Section V discusses privacy-preserving learning, that is, using PPMs in combination with ML, DL, and FL. Section V is further subdivided into subsections of nonFL-based privacy preservation learning, privacy preservation-FL, privacy preservation-FL for IDS, and last but not least, privacy preservation-FL for IoT systems. The purpose of Section V is to review the literature available on the topics mentioned above and to give the readers a detailed review of the research work already done and published in related areas related to the use of FL to preserve privacy for the implementation of IDS in IoT systems. Section VI elaborates on some practical and feasible case studies on using PPFL to implement IDS in IoT systems. Section VI gives specific examples implemented and referenced in the literature. Section VII discusses future research directions and open research questions encountered during the formulation of the survey for the review of the literature on PPFL and its application for the implementation

TABLE 3. Comparisons of referred surveys, reviews, tutorials, and thesis.

Ref.	Paper Type	PPFL	IDS	IoT	Contribution
[3]	Survey/Book	Yes	No	No	Studies and Surveys FL and allied topics in details.
[4]	Survey	Yes	No	No	Studies and Surveys PPFL and its allied topics in detail with a 5W privacy leakage risk scenario.
[5]	Survey	No	No	No	Studies and Surveys the concepts related to security, threat, and vulnerabilities in FL and the applications of FL related to SMC and MEC, trends in attack scenarios, and open research problems.
[6]	Survey	Yes	No	No	Surveys and Studies in detail FL, its applications, types, vulnerabilities, defense techniques, and privacy issues and their mitigation.
[7]	Review	No	Yes	Yes	It reviews the various techniques related to FL and the algorithms that can be used to implement IDS in IoT environments, focusing especially on NIDS.
[8]	Review	No	Yes	No	Does a comparative review of all the IDS techniques and algorithms based on FL and non-IID data sets; it also discusses the limitations and advantages in the available literature.
[9]	Review	No	No	No	It is a review paper that reviews various attack vectors affecting FL and its mitigation techniques and the challenges in FL security.
[10]	Review	Yes	No	No	It studies and reviews FL from a client's perspective. It analyzes if a solution applies to one type of problem, will it also apply to another, and if there are other impacts to consider. It also discusses other types of unsolved client-side FL challenges.
[13]	Survey	No	Yes	Yes	It presents a comprehensive survey on the applications of FL for IoT systems. This includes data sharing in IoT, offloading and caching, detecting attacks, localization, mobile crowd-sensing, and IoT privacy and IoT security.
[14]	Survey	No	No	Yes	It surveys the diverse applications of FL in IoT systems. It evaluates the FL taxonomy on the following metrics: scalability, quantization, robustness, sparcification, security, and privacy. It also lays out open challenges and opportunities related to the topic.
[15]	Survey	Yes	No	Yes	It surveys and studies the applications of FL in IoMT and the healthcare industry. The use of FL to protect the privacy and security of medical data on medical devices is shown highest importance here.
[16]	Survey	Yes	No	No	This is a survey that gives an in-depth study of FL and its applications as per the European-Union GDPR rules and regulations.
[17]	Survey	Yes	No	No	This is a comprehensive survey for the applications of FL in smart-healthcare applications. The domains included but were not limited to healthcare data management, remote health monitoring, medical imaging, and COVID-19 detection.
[18]	Survey	Yes	Yes	No	This survey is an in-depth study of IDS developed using FL in various domains. It gives a broad view of the various techniques and methods utilized in implementing IDS using FL.
[20]	Review	No	No	No	This paper is a state-of-the-art study on the schemes of secure and privacy-preserving medical data sharing with a focus on blockchain-aligned approaches.
[21]	Survey	No	No	No	This is a survey of the methods and applications of PPML rather than PPFL; this survey is meant for readers with normal technical capability rather than a higher level of technical capability.
[22]	Survey	Yes	No	No	This survey discusses the amalgamation of blockchain with FL in applications that require trust, privacy, and security as an essential component. A 3-layer blockchain-based FL system is proposed and reviewed.

TABLE 3. *(Continued.)* **Comparisons of referred surveys, reviews, tutorials, and thesis.**

[23]	Survey	Yes	Yes	No	This survey discusses various attack vectors that affect FL, and it also studies several defense techniques to mitigate these attack vectors in FL and preserve the privacy of client's data and data models used for training.
[25]	Survey	No	Yes	Yes	This survey studies techniques and methodology for implementing IDS and IPS for IoT systems using ML and DL techniques. It also discusses the various IDS and IPS techniques and mitigation approaches to detect and prevent attack vectors on IoTs using ML and DL.
[27]	Survey	Yes	No	No	This survey gives an in-depth study on privacy-preserving aggregation protocols and methods for enhancing privacy and security in FL applications.
[28]	Survey	Yes	No	No	This survey paper talks about the PPDL approaches that can be used in MLaaS applications. A comparison between the state-of-the-art PPDL approaches is also made. An adversarial model is also classified using PPDL to highlight possible PPDL attacks plus mitigation strategies.
[30]	Survey	Yes	Yes	Yes	This survey paper is about the threats and attacks faced by IoT systems. It discusses how FL can be used to detect attacks and malware in IoT systems. It also discusses how FL can be used adversarially in FL systems.
[32]	Review	No	Yes	No	This short review paper discusses the various DL techniques combined with privacy-preserving technologies and their application in various areas like anomaly detection, NIDS, general-IDS among other things. The paper also gives future researchers some pointers for further research with open-research problems in the discussed research areas.
[33]	Thesis	No	Yes	Yes	This Ph.D. thesis focuses on methods of Federated-DL for solutions to botnet-attack-vector detection and mitigation in an IoT-based critical infrastructure.
[34]	Thesis	No	Yes	Yes	This Ph.D. thesis solves the problem of solving cyber-threats in smart IoT systems using FL and Generative Adversarial Networks.
[59]	Survey	No	Yes	Yes	This study is about the application of DL to handle privacy preservation issues in 5G Heterogeneous-Nets, targeting heterogeneous Radio-Access-Networks (RANs), beyond-RAN networks, and end-to-end networks slice.
[139]	Survey	Yes	No	Yes	This survey deals with Privacy Enhancing Technologies employed with FL in the IoHT domain and identifies open research problems in this domain for future research.
[148]	Survey	No	Yes	Yes	This survey discusses IoT security threats from hardware, software, and data in transit perspectives, emphasizing precautions against various threats. It also reviews current security solutions and explores hardware security solutions for IoT-constrained devices. It also examines the impact of emerging technologies like ML and blockchain on IoT security and proposes mitigation strategies.
[153]	Survey	Yes	No	No	This paper provides an in-depth and state-of-the-art overview of the current state of FL attacks and mitigations. It highlights the importance of considering the impact, budget, and scale of FL attacks. Finally, case studies and future research directions toward improving the security and privacy of FL systems are discussed.

TABLE 3. *(Continued.)* **Comparisons of referred surveys, reviews, tutorials, and thesis.**

[165]	Survey	No	Yes	Yes	This survey provides an in-depth study on the use of blockchains and FL to implement IDS and IPS techniques for Industrial IoT systems.
[184]	Survey	Yes	Yes	No	The paper provides a practical tutorial and survey on FL, covering its foundational components, implementation guidelines, exemplary case studies and emerging research trends. It aims to be a comprehensive resource for researchers, developers, and data scientists exploring FL's capabilities in real-world applications.
[185]	Survey	No	Yes	Yes	This paper offers a comprehensive taxonomy of FL-enabled IDS approaches, covering key ML models, datasets, aggregation functions, and implementation libraries from 2018 to 2022. The paper also critically examines the current state of research, identifying key challenges and future directions.
[186]	Survey	Yes	No	Yes	This paper provides FL researchers interested in in-depth and up-to-date exploration of multifaceted FL in a privacy-centric era. This paper is structured with three main objectives. First, within the realm of taxonomies for FL, the study aims to identify and categorize the existing frameworks used in FL research. Second, the study explores the applications of FL by scrutinizing and classifying various use cases in various domains and industries. Third, the study addresses the inherent challenges of FL by examining the associated limitations. This paper serves as a road map towards establishing sustainable massive IoT networks. In addition, it paves the way for interested researchers to pursue and advance the vision of the evolving role of FL in IoT.
This Survey	Survey	Yes	Yes	Yes	This survey examines the processes, techniques and literature to implement IDS in IoT environments using PPFL. Fills a gap in the available literature by focusing on PPFL techniques for implementing IDS in IoT environments.

of IDS for IoT systems. Section VII provides the reader with recommendations for addressing the open research issues identified based on the literature review completed for this survey. Finally, Section VIII concludes the survey by describing the salient features of this survey paper.

II. PRIVACY PRESERVING MECHANISMS

Privacy-preserving mechanisms (PPM) or privacy-preserving technologies (PPT) are capable of providing comprehensive data protection capabilities that minimize the use of personal data of consumers or clients and maximize data security and privacy while enabling market forces and government agencies to gain insights, analyze the data, and optimize their reach as per their needs. Modern IoT devices employ data sharing with many other devices, and these data are then analyzed using AI to make predictions. However, there is a trade-off in data sharing in AI training and predictions, which means that the prediction power is directly proportional to the risk to privacy of the data shared among IoT devices.

Many countries and regions have framed specialized laws to protect citizens' private and sensitive data from the clutches of marketers and hackers. The European Union (EU) has the GDPR to protect citizens' sensitive and private data. In contrast, the US has HIPAA, which protects health-related data from the clutches of health insurance companies, medical companies, and hackers. Similarly, various data protection laws exist in other countries and regions, and some are formulating such laws. Some privacy-preserving technologies are FL (model sharing, not data sharing), anonymization, SMC, HE, DP, blockchain, and TEEs. We will discuss them individually in a comprehensive manner and also discuss how they are used in the recent literature [172], [173], [174].

When applied in conjunction with AI/ML algorithms and techniques, PPM must be able to provide two types of guarantee under the object-oriented privacy guarantee principle: data-oriented privacy guarantee and model-oriented privacy guarantee. The former means that in any PPML solution, a malicious agent must not be able to learn any private information directly from the training/inference data samples or its associated confidential information with the pattern of a specified user. However, the latter means that in any PPML solution, any malicious agent should be unable to deduce any private information from a given model by sending a query to the server multiple times [26].

A. DIFFERENTIAL PRIVACY

DP is a mathematical formulation for privacy-preserving data analysis. The purpose of this project is to ensure

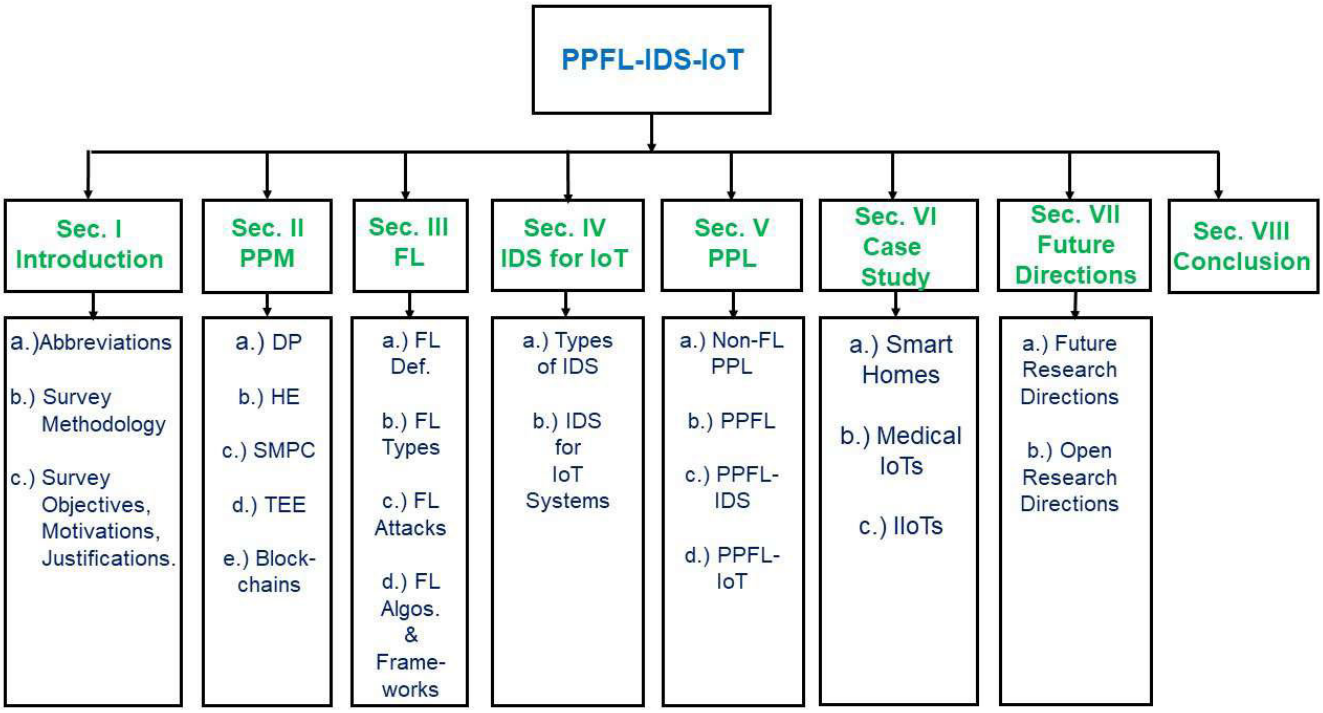


FIGURE 1. Survey structure diagram.

the privacy of individuals in a dataset by preventing the extraction of sensitive information through data analysis. This is achieved by adding noise or random perturbations to the data, making it difficult to identify individual data points. DP has become widely used in data privacy and security, especially in large-scale data collection and analysis by government agencies or tech companies. It has also been used to develop PPML algorithms and statistical analysis techniques. DP has become essential to ensure the privacy of personal information and to ensure that individuals are protected against misuse [171]. DP can be defined as a mechanism M [36], [37], [38], which is considered deferentially private if for any two adjacent databases D and D', and for all possible outputs $S \subseteq R$, it satisfies ϵ-DP when

$$\frac{P[M(D) \in S]}{P[M(D') \in S]} \leq \exp(\epsilon) \tag{1}$$

DP can be classified into local DP and global DP. In local DP, users share and add their data to their database. However, in the case of database administrators, this method is not as trustworthy. Here, noise can be added before appending to the database. In the case of global DP, the database administrator adds noise to the processed data available in the databases. If the database administrator is not trustworthy, privacy will be leaked. The noise is added in global DP before extracting the database information. For global DP, a trusted curator is made the owner of the respective database on which the global DP is applied. The curator is trusted to use the global DP correctly [36], [37]. Figure 2 illustrates the basic mechanism of DP. Figure 3 shows the global and local differences in DP.

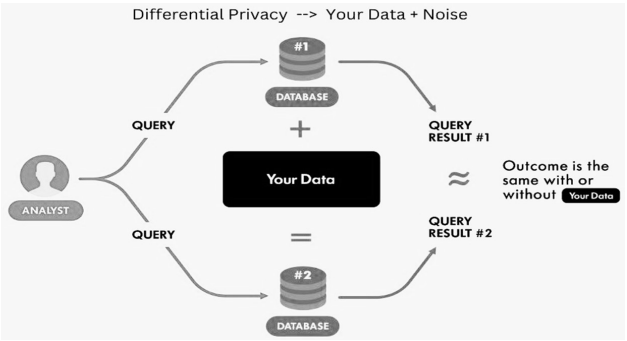


FIGURE 2. Basic mechanism for differential privacy.

Equation 1 summarizes the scenario for the case of global DP, while Equation 2 shows the scenario for local DP.

$$\frac{P[M_n(D_n)\epsilon S_n]}{P[M_n(D'_n)\epsilon S_n]} \leq \exp \epsilon_n \tag{2}$$

Equation 2 holds for any D_n and D'_n where n means the nth participant.

B. HOMOMORPHIC ENCRYPTION

HE is a type of encryption that allows the computation of encrypted data without first decrypting it. This means sensitive information can remain encrypted throughout the computation process, increasing security and privacy. Two main types of homomorphic encryption are fully

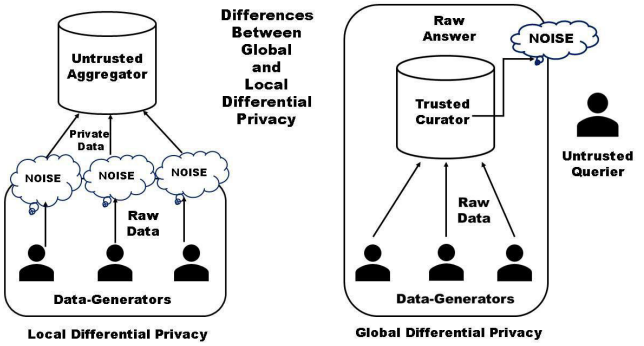


FIGURE 3. Differences between global and local DP.

homomorphic encryption (FHE), which allows for arbitrary computations to be performed on encrypted data, and somewhat homomorphic encryption (SHE), which supports a limited number of computations. The applications of homomorphic encryption include cloud computing, secure data sharing, and secure multiparty computation. However, homomorphic encryption is still an emerging technology with challenges such as high computational cost and limited efficiency and performance [171].

C. SECURE MULTI-PARTY COMPUTATION

SMPC is a cryptographic mathematical formulation that enables many parties to compute a function together over their inputs while keeping their inputs private. In SMPC, the inputs are split and distributed among the parties, and the computation is performed on the split shares without revealing their actual values. The output is also generated in the shares, which can be combined to obtain the final result. This way, SMPC allows for secure collaboration on sensitive data, as the inputs are kept confidential, and the computation results are accurate. SMPC finds applications in various domains, such as finance, healthcare, and online voting, where privacy and security are critical concerns [171].

D. TRUSTED EXECUTION ENVIRONMENT

A TEE is a secure computing device area in a system that ensures that code execution can be trusted. A TEE provides a safe and secure environment for sensitive data processing, such as online banking transactions and personal data storage. The TEE is separated from the central operating system and is protected by hardware security mechanisms, making it difficult for attackers to access sensitive data. TEEs are typically used to protect against software-based attacks and secure the execution of third-party applications. Various certification bodies verify the security of TEEs, and the technology is commonly used in industries such as finance, healthcare, defense, policing, and government [171]. In [176], the authors introduce the concept of Ve-H (verifiable hardware). Ve-H employs a non-intrusive trusted execution environment (TEE) for digest signing, monotonic counters, and trusted timestamps, supporting data notarization and lineage applications [176].

E. BLOCKCHAINS

Blockchain is a privacy-preserving technology that uses cryptographic techniques to secure and validate transactions. Blockchain technology is decentralized and distributed, ensuring that data is secure and protected from unauthorized access or manipulation. In a blockchain, transactions are grouped as blocks, and each block is linked to the previous one through cryptography, creating a secure chain of blocks resistant to tampering. Cryptographic algorithms and encryption protocols in blockchain ensure that data is kept confidential and personal information is protected. Moreover, blockchain eliminates the need for intermediaries and reduces the risk of data breaches and cyber-attacks, further enhancing privacy. As the world becomes increasingly data-driven, blockchain is poised to play an essential role in preserving the confidentiality and security of information, making it ideal for a wide range of applications, such as digital identities, health records, financial transactions, and many more [171].

Within the blockchain domain, numerous applications deployed on permissioned blockchains have been observed to not capitalize on the decentralized blockchain architecture, thus impairing system performance due to low throughput, high latency, and substantial storage overhead. In response to these limitations, LedgerDB, an innovative centralized ledger database introduced by Alibaba Cloud, emulates essential blockchain attributes such as tamper-evidence and non-repudiation while delivering superior auditability and performance. Notably, LedgerDB utilizes a time-stamp authority (TSA) based two-way peg protocol to fortify against malicious activities by both users and service providers, and it supports verifiable data removals, thus aligning with regulatory requirements and optimizing storage efficiency. Empirical evaluations indicate that LedgerDB outperforms contemporary permissioned blockchains, such as Hyperledger Fabric, by achieving an astounding eighty-fold increase in throughput. Consequently, numerous blockchain clients within sectors such as intellectual property protection and supply chain management have switched to LedgerDB, drawn by its exceptional throughput, minimal latency, robust auditability and user-friendly attributes [175].

In recent years, researchers have developed blockchain-like ledger databases as efficient alternatives to permission-based blockchains. Conventional ledger databases, which depend on authenticated structures such as Merkle trees and transparency logs for auditability, face performance issues. In contrast, VeDB (Verifiable Databases) is designed as a high performance verifiable database management system with both verifiable software (Ve-S) and verifiable hardware (Ve-H) components, offering rigorous auditability and broader application scope. Ve-S introduces a novel verifiable shrub array (VSA) with two-layer ordinals, outperforming traditional Merkle tree models due to lower CPU and input-output costs. Improve auditability with efficient timestamp range authentication and fine-grained data verification [176].

Adopting blockchains in various implementations of innovative technologies can never guarantee robust privacy.

There are numerous instances in which blockchains might fall victim to various attack vectors. Existing permissioned blockchains and centralized ledger databases face limitations and threats to privacy, particularly regarding verifiability and high verification costs due to the continuously growing immutable storage [177].

In [177], the authors introduce LedgerDB. The centralized ledger database ensures strong external auditability and rapid verification through a novel concept called Dasein Verification, which incorporates the validation factors of 'what', 'when', and 'who' for formal ledger auditing. LedgerDB employs a fractal accumulating model for faster existence verification and a CM-Tree for efficient lineage verification, along with support for verifiable data mutations. The authors also address vulnerabilities in current time-pegging protocols with a proposed time notary protocol. Evaluations indicate that LedgerDB significantly outperforms traditional methods, achieving notably higher throughput and lower latency than Hyperledger Fabric and QLDB (Quantum Ledger Database) in various applications.

III. FEDERATED LEARNING

Here, we will present the definition of FL and the different types of FL used in applications involving various IoT devices. This section also gives the reader an idea about different attack scenarios malicious actors and adversaries can use to target an FL system. We also compare open-source FL implementation frameworks and various FL aggregation algorithm strategies.

A. WHAT IS FL?

FL is a distributed ML technique that enables training ML models on many decentralized devices without sharing the underlying data. In FL, the ML model is trained on each device using its local data, and only the updated model parameters are shared with a central server for aggregation. This process is repeated iteratively and the model is refined on each device based on its local data. FL allows for PPML, as the local data on each device is not shared with the central server or other devices. It benefits applications where data privacy is critical, such as healthcare, finance, or social networks. FL also reduces the amount of data transmitted between devices and the central server, making it more efficient than traditional centralized ML techniques [171]. Figure 4 shows the general architecture of FL and how it works. Mathematically speaking, FL is well suited for ML models that minimize some objective function (loss function) on a training data set for parameter estimation, particularly for gradient-based optimization algorithms. The minimization objective can be expressed as follows [16].

$$\min_{\omega \in \Re^d} f(\omega) = \frac{1}{n} \sum_{i=1}^{n} f_i(\omega) \tag{3}$$

In this equation, we have the training data in the form of set input-output operators (x_i, y_i), where $x_i, y_i \in \mathbb{R}^d$ for all $i \in \{1, 2, \ldots, n\}$ and n is the number of samples in the dataset. $\omega \in \mathbb{R}^d$ is the parameter vector and $f_i(\omega)$ is a loss function [16].

B. TYPES OF FL

FL can be classified into five essential types, the details of which are listed below [14], [23].

- *Horizontal Federated Learning (HFL)*: HFL is suitable for datasets featuring identical attributes but is distributed across distinct devices. It is divided into two primary categories: HFL for business-to-business applications and HFL for business-to-consumer applications. Conventionally, the former involves a limited cohort of participants, carefully selected during the Federated Learning (FL) training process, typically characterized by devices endowed with robust processing capabilities and technical advantages. Conversely, in horizontal federated learning targeting consumers, the scope widens to encompass hundreds of millions of devices, although only a subset undergoes training in each iterative cycle. [23].
- *Vertical Federated Learning (VFL)*: Numerous vertical federations utilize diverse feature sets. Additionally, the abbreviation VFL denotes heterogeneous federated learning. VFL pertains to scenarios where two datasets share identical sample ID spaces but diverge in their feature set spaces, implying that although the datasets are identical in samples, they possess distinct characteristics. Primarily serving corporate entities, VFL shares similarities with participants engaged in horizontal-to-business federated learning [23].
- *Federated Transfer Learning (FTL)*: FTL shares notable similarities with transfer learning within the traditional machine learning paradigm, a technique commonly employed to enhance a pre-trained model with new features. However, FTL entails a more comprehensive approach involving intermediate learning steps to map to a shared feature subset. This stands in contrast to convolutional neural network (CNN) transfer techniques, which typically include discarding the last few layers of a trained model and fine-tuning it to recognize labels on a limited dataset. An illustrative example of FTL application could extend the scope of VFL to incorporate additional sample cases not universally available across all collaborating institutions [23].
- *Cross-Device Federated Learning (CDFL)*: CDFL finds application in scenarios involving numerous learning devices. Specifically, it pertains to situations where learning occurs across multiple user devices utilizing data generated by individual users, thereby emphasizing cross-device configurations. The inception of CDFL can be traced back to Google's utilization of G-Board user data from smartphones operating on the Android system to construct a next-word prediction model. Key strategies for facilitating CDFL include the selection of clients and implementing incentive designs [23].

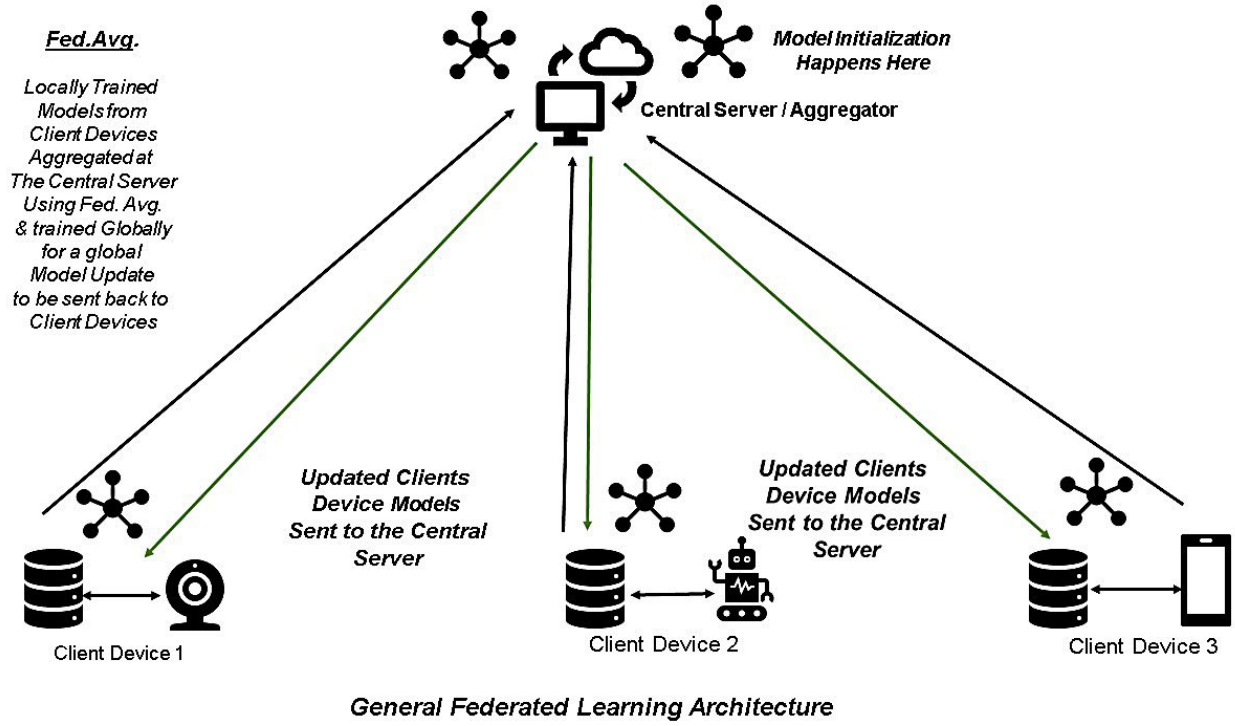


FIGURE 4. General architecture of federated learning.

- *Cross-Silo Federated Learning (CSFL)*: When the number of participating machines is severely limited, all devices become accessible for all rounds, prompting the deployment of CSFL in such cases. The training framework and architecture of CSFL exhibit significant deviations from conventional approaches. In this context, user devices can share specific intermediate outcomes, in addition to model parameters, contingent upon the intricacies of the training process. This collaborative sharing facilitates gradient calculations for other parties, with the involvement of a central server, if at all, acting as a neutral intermediary in the procedure. The training data may adopt either a HFL or VFL format. CSFL finds widespread usage, particularly in scenarios involving multiple large corporations [23].

From the perspective of networking structure, FL can be divided into two main types: centralized FL and decentralized FL. Figure 5 shows the differences between the two.

- *Centralized Federated Learning*: This architecture stands as one of the most prevalent frameworks for Federated Learning (FL), centered around a pivotal role played by a central server in conjunction with a fleet of devices tasked with training an FL model. Each device independently trains a network model using its respective data set in a single training round. Subsequently, all devices transmit their trained parameters to the central server, which employs a weighted averaging algorithm, such as Federated Averaging (FedAvg), to aggregate these parameters. The resultant global model is then distributed to all devices for the subsequent round of training. Following completion of training, each device attains an identical global model, aligned with its locally trained model. In centralized FL, the server assumes a central role in orchestrating the aggregation and dissemination of model updates to the devices, thereby facilitating the achievement of the FL task at hand [14].

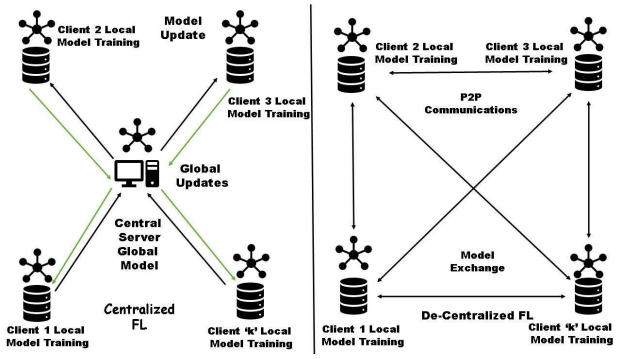


FIGURE 5. FL types as per network architecture.

- *Decentralized Federated Learning*: On the other hand, decentralized FL is a network architecture without any centralized server to coordinate the model training process. All client devices are connected in a decentralized peer-to-peer (P2P) structure to perform model training, as shown in Figure 5. Each client device conducts local training using its data sets. In the next step, each client device implements model aggregation using the model

updates received from their neighboring client devices through P2P communication to reach a consensus on the global model update. Decentralized FL came into being to replace centralized FL when the scenario does not allow for there to be a communications server or the available network topology is highly scalable. Due to its highly contemporary features, decentralized FL can be integrated with blockchain technology. Client devices can communicate through blockchain ledgers, where model updates can then be offloaded to the blockchain for a secure model communication exchange and aggregation [14].

C. TYPES OF ATTACKS ON FL

According to the literature published in the last three years, this subsection introduces common attack vectors that plague FL and several ways to mitigate them. Figure 6 shows the main attack vectors that affect the FL paradigm.

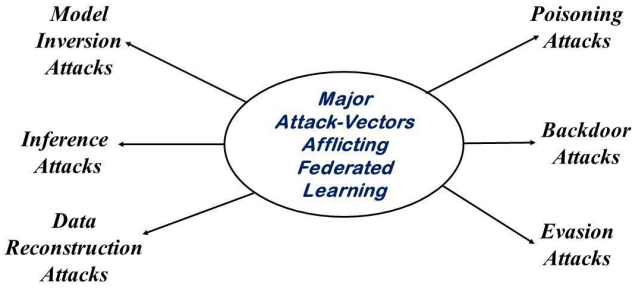


FIGURE 6. Major attack vectors afflicting FL.

- *Backdoor Attacks*: Compromised machines participating in FL can establish a backdoor by training the model on specified backdoor data. These attacks can be dangerous because they can produce false positive results with high accuracy. Some highly technical backdoor attacks can keep the backdoor alive and well during training so that it can never be forgotten. Backdoor attacks have many use cases but always damage the FL training architecture and consensus mechanism. Backdoor attacks are often referred to as targeted assault attacks. One of the mitigation mechanisms for backdoor attacks is to include a strong DP mechanism in FL. In this scenario, DP ensures that malicious actors cannot be able to reverse engineer the data parcels to identify individual clients, even if they have access to the data containing such information [23].
- *Inference Attacks*: Inference attacks are defined as a data-mining technique in which data are inferred from a set of partial information. The attacker analyzes large amounts of data and can illegally gain knowledge about a database or its content without access. The significance of inferred data lies in its categorization as trivial, which occurs under the specific condition that no meaningful insights or access to additional information can be derived from it. In other words, the data is deemed inconsequential and lacking in valuable content if it fails to yield meaningful inferences or allow for further exploration and understanding of the underlying subject matter. However, the inference attack is successful if an attacker can piece together valuable information that should be protected at the highest level. Generative Adversarial Networks (GANs) can conduct an inference attack, namely a model inference attack. In the literature, several ways can mitigate inference attackers in the FL. Some mitigations use the following privacy-preserving mechanisms: (i) homomorphic encryption, (ii) trusted execution environments, (iii) differential privacy, and (iv) secure multiparty computation [23].
- *Poisoning Attacks*: Poisoning attacks in the FL scenario incorporate two main types: (i) data poisoning and (ii) model poisoning. In data poisoning, the attacker breaks into the FL database and introduces false and misleading data that corrupts the whole FL model. The attack can be classified into two categories: a) clean label attacks and b) dirty label attacks. The former assumes that the attackers cannot alter the training data labels. In the latter, the attacker can add a modifier to multiple data samples, and with that, the attacker will be able to classify the labels in the training database wrongly. In the case of model poisoning attacks, the attacker does not manipulate the training data but plays with the model parameters and weights. The attack can affect the model performance without getting noticed by the FL system. Since FL works in a distributed manner, the attacker targets individual models and poisons them in some form. The primary motivation for implementing the attack is that all the client devices in FL have direct access to the global model. This gives the malicious clients access to poison the weights and parameters of the global model using the weights and parameters of the local model. Many researchers have shown that model poisoning attacks are more promising in FL than simple ML/DL [35]. The primary defense mechanisms for data poisoning attacks are data sanitization, influence-based defenses, and the stochastic gradient descent (SGD) algorithm. On the other hand, the mitigation to model poisoning attacks is the use of byzantine robust aggregation, clustering-based strategies, principal component analysis (PCA), and the *Romoa* algorithm, which improves fairness among the participating clients and promotes more robust attack detection [23].
- *Model Inversion Attacks*: The occurrence of attacks arises during training sessions conducted on a supervised neural network, particularly when a non-public dataset is segregated from the general dataset. Simply put, inversion attacks entail efforts by malicious entities to extract the dataset. The model inversion problem can be conceptualized as a variational inference problem, offering a more nuanced statistical perspective. Armed with access to the original training dataset and additional secretive data, a malicious actor may execute

such an attack to retrieve specific variables from the training set. Given the interconnected nature of variables between the existing training data and confidential data, the resulting individual dataset incorporates variables from both sources. Data retrieved from the training dataset tends to exhibit greater accuracy compared to characteristics inferred from it, owing to the potential for error and prediction inaccuracies. Mitigation strategies for such attacks in Federated Learning (FL) are not straightforward. Nonetheless, the literature suggests that a combination of model inversion attack detection techniques, integrated with Generative Adversarial Networks (GANs), Deep-Convolution GANs, and Principal Component Analysis (PCA), proves effective in addressing this challenge [23].

- *Data Reconstruction Attacks*: In straightforward terms, attacks encompass any method aimed at reconstructing or piecing together a private dataset solely based on publicly available information about the said dataset. Protection of the private dataset is paramount because it includes personally identifiable information, which must remain shielded from potential attackers. Attackers may lack direct access to the private dataset but could obtain access to publicly accessible aggregated statistics. These aggregated statistics may not necessarily be pristine, as they might contain noise or unclean data. Yet, they remain susceptible to exploitation by attackers for reconstructing or assembling a substantial portion of the private dataset, provided the statistics remain coherent and uncorrupted. It is worth noting that identifying data reconstruction attacks spurred the initial studies on Differential Privacy (DP). Within the context of Federated Learning (FL), the distributed nature of the attack poses significant challenges, given the involvement of numerous clients, any of which could potentially act dishonestly and execute such attacks. Mitigation strategies against these attacks include enforcing safety thresholds for gradient-based data reconstruction, achieved through a microscopic analysis of neural networks utilizing rectified linear units (ReLUs) as activation functions. Additionally, a system founded on adversarial training, comprising three modules—malicious reconstruction, noise regularization, and distance correlation minimization—provides a highly effective defense against such attacks [23].
- *Evasion Attacks*: Evasion attacks involve adversaries manipulating the information tests conducted on the target model to evade detection. These attacks manifest in diverse forms, with some employing modified tests that superficially resemble the originally designed information tests but are, in reality, purposefully crafted to deceive the classifier. Within Federated Learning (FL) settings, evasion attacks are not uncommon, showcasing specific vulnerabilities distinct from other Machine Learning (ML) or Deep Learning (DL) systems. In white-box scenarios, adversaries employ limited optimization techniques, generating perturbations such as projected gradient ascent and loss function maximization with norm constraints. Conversely, in black-box scenarios, adversaries substitute models trained on comparable data. FL's characteristic of granting attackers access to models and local training loss functions renders it susceptible to evasion attacks. Several mitigation techniques have been proposed to counter these attacks, including adversarial training, randomization-based defenses, and noise removal [23].

D. FL ALGORITHMS AND FRAMEWORKS

In this subsection, we discuss various algorithms used for learning and aggregation functionality along with FL to implement and optimize the whole FL process. We also compare the implementation of various FL averaging algorithms in the literature. We introduce various open-source frameworks and application programming interfaces (APIs) available to researchers and programmers to implement FL for various purposes.

In federated learning, many algorithms provide aggregation functionality for local client models to aggregate at the central server to produce an updated global model. The most common among them is the federated averaging algorithm (FedAvg), developed by Google in 2017. FedAvg performs the weighted average of the collection of local client models by computing the weighted average of the arrays of parameters. Each array of parameters corresponds to a local client model, and the weight factor is the proportion of the client data volume to the total data volume. More complex variants of FedAvg have been designed to alleviate the difficulties associated with heterogeneous non-IID datasets [184], [185].

In terms of state-of-the-art literature, the other major variants of FedAvg are FedPAQ, FedMA, FedProx, and FedPlus. FedPAQ has an equivalent aggregation approach to FedAvg while providing a higher level of optimization in environments with certain computing and network constraints. FedPAQ appends three main features to FedAvg, viz., periodic averaging, partial device participation, and quantized message passing. FedMA (Federated Matched Averaging) is a federated learning approach that addresses the issue of permutation invariance of neural network parameters. Unlike the conventional FedAvg method, which performs coordinate-wise averaging of model parameters, FedMA employs the principles of Probabilistic Federated Neural Matching (PFNM). This involves a matching process of the clients' neural networks, enabling layer-wise averaging of the model parameters. By accounting for the permutation invariance property, FedMA aims to improve the accuracy and convergence of federated learning compared to directly averaging the parameters across clients. FedProx (Federated Proximal) is a variant of the proposed FedAvg algorithm to address the system and the statistical heterogeneity prevalent in FL settings. The key objective of FedProx is to mitigate the effects of heterogeneity among clients by introducing a

proximal term that restricts local models from significantly deviating from the global model by limiting the impact of local updates. Consequently, FedProx encourages local models to remain closer to the central model during the training process. Additionally, FedProx incorporates mechanisms to handle device heterogeneity, allowing it to adapt to scenarios involving clients with varying computational capabilities. By accounting for both system and statistical heterogeneity, FedProx aims to improve the performance and convergence of federated learning in practical, non-IID settings. Fed+ (or Fedplus) constitutes a suite of algorithms designed to address data heterogeneity, particularly in the context of non-IID data. The method extends the conventional Federated Averaging (FedAvg) algorithm by incorporating a penalty function applied to each client. The modification alleviates the constraint that requires the convergence of all client weights to a single point, thereby enhancing the robustness of the algorithm in handling diverse data distributions between clients [185].

In the above paragraph, we discussed widely considered approaches in FL settings for various use cases and contextual scenarios. A significant number of other FL aggregation functions have also been defined in the referred literature. Some of the secondary FL aggregation algorithms and strategies presented by researchers and developers for various use cases are discussed here individually.

The turbo-aggregate technique originates from the concept of enhancing the security of communications between clients and servers by employing secure aggregation techniques. Analogous to FedAvg, FedCD initiates with a global model hosted on a centralized server, which is subsequently distributed to all participating devices. During each iteration, clients train distinct models and compute a performance-related score for these models. To address non-IID data scenarios, FedCD employs a strategy where, starting with a single model in certain rounds, it clones high-performing models based on their scores and eliminates those with lower scores. Ultimately, the model with the highest score is selected for continuation. Nonetheless, this cloning procedure can strain the devices' resources, potentially causing delays in the overall process. Krum FL aggregation is intended to handle scenarios with several malicious peers. It selects specific weights from the clients' weights, aiming to minimize the distance to their closest neighbors, excluding a predefined number of potentially malicious clients. The selected weights are then sent to the server in each round. However, a major limitation of Krum aggregation is that it necessitates prior knowledge of the exact number of malicious clients, which often makes it impractical for real-world applications.

Semi-asynchronous Federated Averaging (SAFA) addresses situations marked by poor communication conditions, where certain clients may become disconnected due to varying capabilities. To manage this, SAFA employs an asynchronous aggregation method to lessen the effect of slow-responding clients. Additionally, the authors introduce a cache structure in the cloud to lower communication costs and a post-training client selection strategy to enhance convergence. Concept-Drift-Aware Federated Averaging (CDA-FedAvg) extends the FedAvg algorithm to handle concept drifts, which occur when data patterns change due to specific events or circumstances. In CDA-FedAvg, the server aggregates client weights asynchronously without a predefined sequence for updating the global model. Meanwhile, clients maintain two distinct data storages—short-term and long-term—to monitor for potential drifts. Upon detecting a drift, additional training rounds are conducted to adapt to the changes introduced by the concept drift. SecureBoost is a framework designed for Gradient Boosting Decision Trees (GBDT) on vertically partitioned data, aimed at enhancing client privacy. It facilitates the sharing of a GBDT model across multiple parties while adhering to stringent privacy constraints. SecureBoost employs a privacy-preserving protocol based on entity alignment techniques, the XGBoost framework, and Paillier encryption to determine inter-database intersections. These techniques are utilized by clients to calculate the model weights in each training round before transmitting them to the server. Communication-Mitigated Federated Learning (CMFL) is predicated on the principle of discarding irrelevant updates from certain clients. To determine the relevance of a model update, each client compares its local update with the global model by calculating the percentage of parameters with differing signs (positive or negative). A higher percentage of differing signs indicates a less relevant update, which can be discarded if it exceeds a specified threshold. Dynamic Weighted Federated Averaging (DW-FedAvg) builds upon FedAvg by incorporating mechanisms to incentivize high-performing clients. It adjusts client weights using weight factors, which are initially set based on the size of each client's dataset. Throughout the training, clients whose accuracy exceeds the average are assigned higher weight factor values, encouraging their contributions. Conversely, clients with below-average accuracy receive lower weight factor values to reduce their impact. After adjustment, weight factor values are normalized by dividing each by the sum of all clients' weight factor values. This iterative approach continues across all training rounds to optimize client participation based on their performance [185].

We now introduce the readers to the common FL frameworks and APIs that researchers and developers use in the state-of-the-art literature. Most of the frameworks and APIs are open source and are easily available online for researchers to use in their studies. Some of the common frameworks are the following, as listed and described below.

1) TensorFlow Federated (TFF): A publicly available machine learning platform that extends the capabilities of TensorFlow to support federated learning (FL). TensorFlow is renowned as a leading platform in deep learning, enjoying widespread popularity and robust community support from developers. Consequently, users familiar with TensorFlow will find TensorFlow

Federated (TFF) advantageous for initiating model development in federated learning compared to other platforms [184].

2) Flower (Federated Learning Framework): Flower has garnered recent acclaim for its user-friendly interface and ease of use relative to other platforms. Despite being launched more recently than TensorFlow Federated (TFF), Flower benefits from a dedicated community and active maintenance. Both TFF and Flower share many common characteristics; however, TFF offers partial support for creating non-IID data partitions and provides slightly stronger support for differential privacy (DP). It is noteworthy that TFF allows for the definition of deep learning models compatible with TensorFlow and Keras frameworks. In contrast, Flower also supports models written in PyTorch, a significant tool in the field of deep learning [184].
3) FATE (Federated AI Technology Enabler): Finally, neither TensorFlow Federated (TFF) nor Flower provide suitable solutions for VFL scenarios within their frameworks, whereas FATE does. Among the frameworks that support VFL, FATE stands out for offering the most extensive options and possessing the most comprehensive and clear documentation. Unlike TFF and Flower, which are built on top of TensorFlow, Keras, or PyTorch, FATE's implementation is independent of any prior framework. Instead, it defines its procedures based on a pipeline approach, where various components are added, compiled, and executed. FATE also provides some mechanisms for addressing problems in HFL scenarios; however, it offers significantly fewer options and performs worse than other frameworks. Nevertheless, its applicability in VFL scenarios is excellent. Conversely, when tackling HFL problems, the choice between TFF and Flower depends on various factors. The factors include the user's familiarity with TensorFlow or PyTorch, the necessity of incorporating data privacy mechanisms (TFF includes DP mechanisms, whereas Flower does not), the performance of the models (Flower models generally outperform those of TFF), and other distinguishing characteristics of the frameworks [184].
4) FedML (Federated ML): FedML is an open-source research library and benchmarking tool that accommodates various FL computing paradigms. It supports on-device training for edge devices, distributed computing, and single-machine simulation. Additionally, FedML offers a flexible and generic API design, along with extensive reference implementations of optimizers, models, and datasets. The architecture of FedML incorporates virtual nodes and a central server [182].
5) Substra (FL Framework): Substra is an open-source framework devised to facilitate distributed learning with stringent privacy assurances, ensuring that data remains localized to its respective nodes. Within this network, only predictive models, algorithms, and non-sensitive metadata are exchanged. Substra supports various computational paradigms, including the parallel computation strategy utilized in FL [182].
6) Open FL Framework: OpenFL is an open-source tool implemented in Python, specifically designed for training ML algorithms using the FL paradigm. It supports training pipelines in both TensorFlow and PyTorch and is extensible to other ML frameworks. The significance of OpenFL is underscored by its scalability, emphasis on trusted execution environments, and its ability to seamlessly transition centralized machine learning models to a federated training pipeline [182].
7) PySyft: It is an open-source Python library, a pioneering framework devised to facilitate secure and privacy-preserving DL computations. Developed by the OpenMined community, this innovative library employs cutting-edge techniques such as FL, DP, and SMPC to decouple private data from model training. PySyft seamlessly integrates with renowned DL frameworks like PyTorch and TensorFlow, enabling researchers and practitioners to leverage its capabilities within their existing workflows. The library supports two modes of computation: dynamic computations performed on obfuscated data and static computations, which involve constructing computation graphs that can be executed at a later stage in a different environment, thereby enabling distributed and collaborative training scenarios. To facilitate network communication and FL across diverse devices, such as web and mobile platforms, PySyft employs PyGrid, a complementary framework. PySyft is currently in its beta release stage, signifying its active development and ongoing efforts towards achieving production-ready status [185].
8) PaddleFL: PaddleFL is an open-source Python-based FL framework built on PaddlePaddle, engineered for production environments. It supports HFL where parties hold different data samples, and VFL with disjoint feature subsets across parties while planning future development, including integrating federated transfer learning capabilities to leverage knowledge from source domains under federated constraints. Through its robust implementation, PaddleFL aims to enable privacy-preserving, decentralized ML solutions across diverse applications [185].
9) IBM-FL: It is an open-source, enterprise-grade Python library that streamlines the deployment of FL systems in production, providing a foundational FL layer extensible with advanced features. It supports conventional ML algorithms like linear regression and k-means, deep neural networks (DNNs), and seamless implementation of novel FL algorithms, encompassing supervised, unsupervised, and reinforcement learning (RL) paradigms. Actively maintained, IBM-FL empowers organizations to leverage decentralized, privacy-preserving ML solutions by harnessing FL's potential while ensuring data privacy and security compliance [185].

10) FederatedScope: It is a relatively new, flexible, open-source Python platform for FL designed with an event-driven architecture to manage heterogeneous data, resources, behaviors, and learning goals across research and production environments. It prioritizes usability through multiple programming interfaces and extensibility via seamless integration of new <event, handler> pairs. Despite being actively maintained, a stable release has not yet been designated, given its novelty. FederatedScope aims to drive FL innovation by enabling the development and deployment of solutions that can effectively handle diverse real-world requirements while ensuring data privacy and security [185].

1) PERFORMANCE COMPARISON OF FL AGGREGATION STRATEGIES VIA EXPERIMENTATION

Here, we compare the performance of several FL aggregation algorithms by conducting small experimentation using Google Colab, a Python-based online jupyter-notebook platform for ML and DL modeling, coding, and implementations. We experimented with the TFF and Flower FL frameworks using the TensorFlow MNIST dataset (IID). The experiments were conducted on Google Colab and Python version 3.0 using CPU runtime. We compare the performance of FL on the MNIST dataset for training and evaluation, using accuracy as a metric of two FL averaging strategies, namely FedAvg and FedProx. We used 100 client devices and 20 clients per round for 20 rounds to run FL-FedAvg and FL-FedProx until convergence. From this experiment, it was observed that FL-FedAvg performed a bit better than FL-FedProx. For the FedProx experiments, the hyperparameter μ was taken to be 0.1. μ is a hyperparameter that controls the extent to which local updates should stay close to the global model. The accuracy achieved by FL-FedAvg was 98.90%, while for FL-FedProx, the accuracy was 97.98%.

Figures 7 and 8 display the graphical results in the experiments. In this PPFL survey, we included local DP at the client level with the noise parameter ϵ set to 0.001 and experimented again with the same frameworks, dataset, and FL aggregation strategies. The results after incorporating local DP at the client level indicated that FL-FedAvg performed better than FL-FedProx. The accuracy of FL-FedAvg with DP was 89.98%, and that of FL-FedProx was 87.58%. Figures 9 and 10 display the results in local DP experiments. From the results, we can say with caution that additional PPMs such as DP, HE, SMPC, TEEs, and blockchains with FL will affect the performance of such algorithms or techniques. Last but not least, we tested out the swiftness and efficiency of the Flower FL framework along with Pytorch. In this scenario, we performed a classification task with the MNIST dataset using simple CNN algorithmic modeling. The other parameters used in this task are as follows: the number of clients used was 100, the number of classes was taken to be 10, the number of epochs used was 2, the learning rate was 0.01, the batch size of tests was 25, the FL aggregation strategy was FedAvg, and finally the number of rounds was taken to be 10 rounds per client. The maximum accuracy reached after 10 rounds for this aforementioned experiment was 96.26%. Figure 11 displays the results of the experiments.

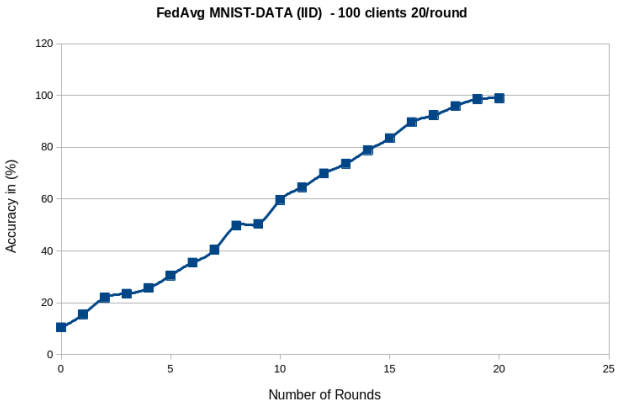


FIGURE 7. FL-FedAvg performance MNIST-IID.

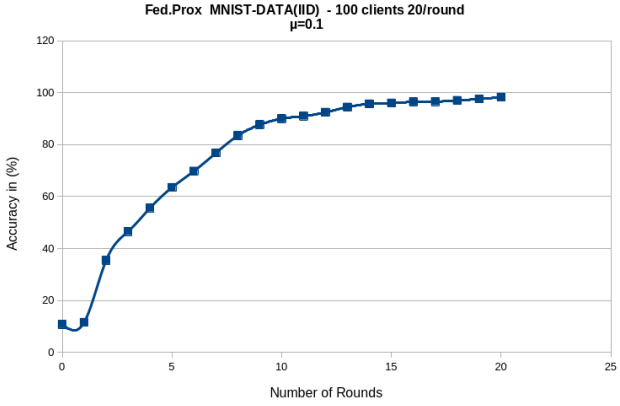


FIGURE 8. FL-FedProx performance MNIST-IID.

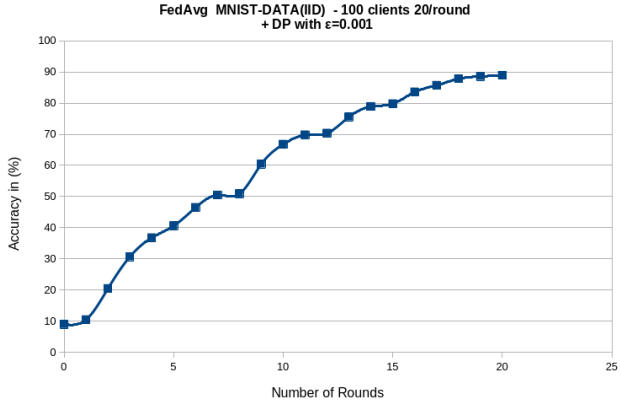


FIGURE 9. FL-FedAvg with DP performance MNIST-IID.

2) PERFORMANCE COMPARISON OF FL AGGREGATION STRATEGIES AND FL FRAMEWORKS

Here, we refer to the recent literature on PPFL and allied topics where researchers have experimented and compared

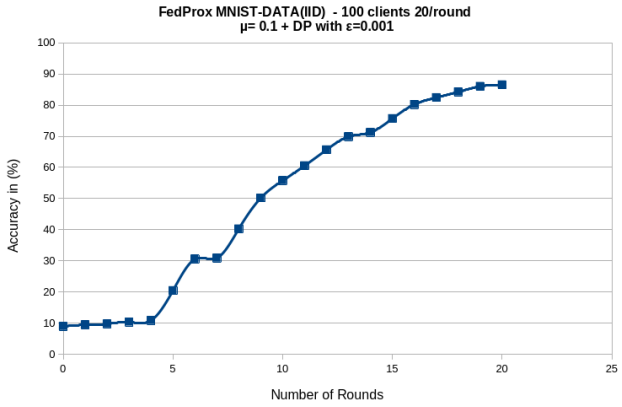


FIGURE 10. FL-FedProx with DP performance MNIST-IID.

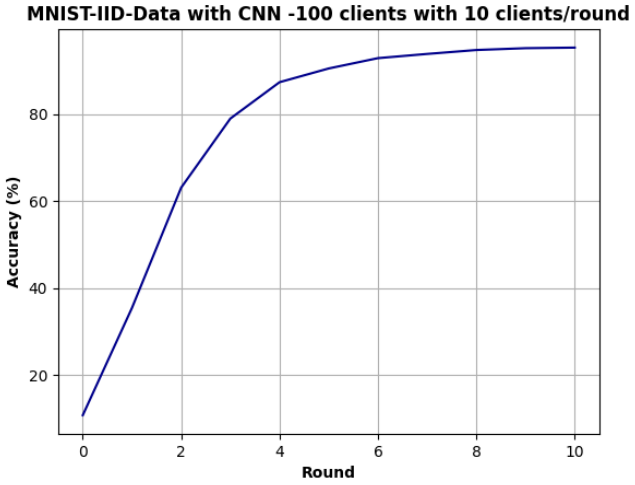


FIGURE 11. Flower-FL with Pytorch + CNN classification performance MNIST-IID.

the performance of various FL aggregation strategies and frameworks in varied scenarios.

Like in [182], the authors experiment and evaluate the performance of leading FL frameworks: Flower, FedML, Substra, and OpenFL, based on metrics including total training time, loss and accuracy values, and CPU and RAM usage with varying client counts. Flower achieves the highest accuracy (98% on average) but suffers from high loss values and significant training overhead. FedML, while demonstrating low accuracy (68% on average) and significant CPU usage (over 95%), shows efficient training time scalability. Substra excels in resource efficiency with low CPU and RAM usage, but faces exponentially increasing training time. OpenFL balances performance with high accuracy (more than 96%) and stable resource usage despite slightly higher training time at low client counts.

In [183], the authors analyze two scenarios that demonstrate the potential of FL for efficient and privacy-preserving machine learning in IoT networks. It begins with the mathematical foundations of key aggregation algorithms in FL, specifically FedAvg and FedProx. Simulations using the Flower framework evaluate the algorithms by training deep neural networks on standard datasets and comparing their accuracy and loss metrics. The results underscore the trade-off between privacy and accuracy through simulations implementing the DP method in the PyTorch and Opacus frameworks, applied to typical FL datasets and data distributions for both the FedAvg and FedProx strategies. The authors in [183] focus on comparing convergence analysis to evaluate aggregation strategies in an FL framework on non-IID data and providing insight into DP in FL techniques and algorithms tailored for IoT networks. The experimental setup for performance comparison in [183] is carefully crafted for model training using heterogeneous unbalanced, non-IID MNIST data sets. The MNIST data set in [183] is divided into 1000 clients. Each client has examples of 2 out of 10 class labels. The number of epochs is 10; the batch size is 10. In the training hyperparameters, the number of clients per round is 10, and the number of rounds is 100. The learning rate is set at 0.03, and the straggler fraction is set at 0.9, which means that the 90% stragglers and the proximal μ are set at 0.0 for FedAvg, and the proximal μ is set at 1 for FedProx. The simulation results in [183] indicate that FedProx achieves a better and more consistent convergence compared to FedAvg, particularly in highly heterogeneous environments. FedProx demonstrates significantly more stable and accurate convergence behavior than FedAvg in [183]. The authors in [183] compare FedAvg without DP and FedAvg with DP to assess the impact of DP on the accuracy of the model. The results highlight the trade-off between improved privacy and performance degradation. The performance difference arises from the core principle of DP, which involves adding unbiased noise to parameter gradients during model updates. This noise addition prevents the model from memorizing the training data in each iteration, while still allowing the learning process to continue. Considering the application of the FedProx algorithm in IoT networks, the authors in [183] incorporated DP into FedProx to evaluate model performance on heterogeneous data distributions. Without DP, the accuracy of the model improves with each round, reaching 0.9146 in the 20th round. With DP, FedProx achieves an accuracy of 83.53% in the 20th round while maintaining privacy.

In [184], the authors compare various scenarios and circumstances in which FL is implemented along with ML and DL algorithms for various tasks. The results of the tasks are described below.

1) Image classification task using DL in HFL, compared for 3-FL frameworks, namely FATE, TFF, and Flower. The final results for this task show that out of the 3-FL frameworks, the best performance was given by the Flower-FL framework. The testing accuracy of Flower when used with CNN in HFL for MNIST image classification was 98.7%, while that of TFF was 96.3%. FATE-FL framework's performance was minuscule when compared to Flower and TFF in this task.

2) Sentiment analysis task using DL in HFL, compared for 2-FL frameworks, namely TFF and Flower. The final results for this task observed the testing accuracy of the Flower-FL frameworks was 96.6%, while that of TFF was 88.6%.
3) Performing VFL classification task using decision trees and FATE. The authors in [184] found that for VFL classification the non-IID partitioned data performed optimally with the use of decision trees and the FATE-FL framework.
4) Implementing DP task with HFL and TFF (IID and non-IID data partitions) to increase the privacy during training process. In this task, the authors observed that the increasing amount of noise factor to introduce privacy during the training process negatively affected the performance. In the observations, for a noise factor of 0.05, the testing accuracy was around 79.98% for non-IID and around 88.90% for IID data partitioning. Similarly, it was observed that, for a noise factor of 0.50, the testing accuracy was around 27.98% for non-IID and around 59.99% for IID data partitioning.
5) Clustering task with k-means using HFL and TFF (IID data partitions-MNIST). In this task, the authors conducted a comparative analysis between the centroids obtained from a traditional centralized k-means algorithm and those derived from a federated implementation involving 10 client nodes. Both algorithms were executed for 10 iterations to ensure a fair comparison. The results demonstrate a notable similarity between the two approaches: a majority of the centroids in both cases delineate distinct digits. However, it is worth noting that certain centroids in both implementations exhibit ambiguity, failing to represent a single, clearly identifiable digit. The ambiguous centroids appear to encompass instances from multiple-digit classes, suggesting a degree of overlap or uncertainty in the clustering process. This observation underscores the inherent challenges in unsupervised learning tasks, particularly when dealing with complex, multi-class datasets in both centralized and distributed computing environments.
6) Clustering task with k-means using VFL and FATE (non-IID data partitions-Credit2-dataset). To validate the efficacy of k-means clustering in a federated learning context, an analysis of the Davies-Bouldin Index (DBI) was conducted. The DBI, a minimized clustering metric, evaluates both the compactness within individual clusters and the inter-cluster distances. The study compared the DBI values obtained from two implementations: a centralized k-means algorithm utilizing all attributes from a single data origin, and a VFL k-means approach employing data distributed across two distinct parties. It is noteworthy that the k-means algorithm in both scenarios was configured to terminate upon reaching convergence or when cluster assignments remained static for a predefined number of iterations. The experimental results reveal that both implementations ultimately achieve the same minimum DBI value, indicative of comparable clustering quality. However, the centralized version exhibited faster convergence, requiring fewer iterations to reach the optimal solution. The findings demonstrate that federated k-means can attain clustering performance on par with its centralized counterpart, even under conditions of restricted data access. This underscores the potential of federated learning approaches to maintain analytical efficacy while adhering to data privacy and segregation requirements.

IV. INTRUSION DETECTION SYSTEMS FOR INTERNET OF THINGS

IDS can detect and react to security threats to standalone and networked systems like IoT devices. As IoT devices are increasingly connected to the Internet, IDS can help identify and prevent unauthorized access, data breaches, and other security incidents by monitoring network traffic and system activities, identifying suspicious patterns or anomalies, and alerting security personnel or automated systems about potential threats. By deploying IDS for IoT devices, organizations can strengthen their security posture and mitigate threats to IoT deployment. Although IDS is not a silver bullet for IoT security, they are essential for holistic security strategies to protect IoT assets and systems.

IDS implements various mechanisms to detect and alert unauthorized access or malicious activities within a network or system. They come in multiple types. The first type of IDS is network-based IDS (NIDS), which monitors network traffic for signs of intrusion. NIDS can detect various attacks, including port scanning, denial of service (DoS), and common vulnerability exploits, and monitor network traffic for suspicious activities and attempts to identify potential intrusions or attacks. The systems typically work by analyzing network packets as they traverse the network and comparing them to known attack signatures or behavioral patterns. NIDS can be deployed at key points in a network, such as the perimeter or within critical subnets, to provide continuous monitoring and detection capabilities. They can also be configured to trigger alerts or take automated actions in response to detected threats.

The second type is the host-based IDS (HIDS), installed on individual hosts or servers, and monitors system activities to detect unusual behavior or suspicious activities. HIDS can monitor the activity and behavior of particular hosts or endpoints on a network. The systems typically run on hosts and collect data from various sources, such as logs, operating system events, and file system changes. HIDS can detect many suspicious activities, such as unauthorized access attempts, malware infection, and configuration changes. They can also be configured to alert security teams or take reactions in response to detected threats. HIDS detects insider threats or attacks from an organization's network. By monitoring individual hosts, HIDS provides a granular view of network

activity that can help security teams identify and respond to potential threats quickly and effectively.

Intrusion Detection Systems (IDS) can be further categorized into behavior-based IDS (BIDS) and anomaly-based IDS (AIDS), offering valuable tools for securing both hosts and networks (HIDS and NIDS). BIDS utilizes machine learning or predefined behavioral patterns to analyze user and application activities. This allows it to identify suspicious deviations that might signal intrusion attempts, even those evading traditional IDS techniques. In contrast, AIDS focuses on network or system behavior, establishing a baseline of normalcy and identifying deviations as potential threats. Although this enables the detection of novel attacks, it can also lead to more false positives. The key distinction lies in how they identify anomalies. BIDS leverages known attack patterns to detect malicious activity, whereas AIDS observes and learns normal behavior, triggering alerts upon deviations. Importantly, AIDS does not rely solely on statistics or ''profiles'' but can incorporate AI models for enhanced anomaly detection. Both BIDS and AIDS offer distinct advantages. BIDS excels at catching known attacks and those mimicking specific behaviors, while AIDS excels at spotting novel threats and unusual patterns. However, both methods require careful tuning to minimize false positives and ensure efficient security monitoring. By combining these approaches within the deployments of HIDS and NIDS, organizations can achieve a layered defense that effectively detects and addresses various security threats.

For IDS applied to various IoT systems, the authors of [74] propose IDS based on Federated Blending, namely F-BIDS. The authors aim to detect the reverse engineering attack on the FL training process in a lightweight manner. The F-BIDS model combines the FL process with the metaclassifier and a blending ensemble for attack classification. F-BIDS, which uses an FL-based DL blending process, supports heterogeneous architectures across local and global models. The model can rapidly analyze a large amount of data for evaluation purposes and uses recent and realistic datasets like *Edge IIoTset* and *InSDN* for a comparative analysis between the F-BIDS and the centralized ML model.

The authors in [66] introduce a distributed cloud storage system employing an IDS powered by the feedforward deep learning neural network (FDLNN) algorithm. The IDS relies on FDLNN, while the data broker utilizes the Lightweight and Efficient Handover (LFEHO) algorithm. Connectivity in the Data Center (DC) is managed through the Message Queuing Telemetry Transport (MQTT) protocol for lightweight devices and the Advanced Message Queuing Protocol (AMQP) for heavy-weight devices. Performance evaluation involves comparing FDLNN with the DLNN and ANN algorithms using the *TON-IoT* data set. The LFEHO algorithm is also compared with the Improved Adaptive Neuro-Fuzzy Inference System (IANFIS) and the Adaptive Neuro-Fuzzy Inference System (ANFIS) for data broker performance. FDLNN achieves an accuracy of 96. 12% for 500 nodes, while LFEHO demonstrates superior performance in data broker analysis.

The authors of [76] use federated mimic learning to implement privacy-preserving IDS for IoT systems. They developed a new technique that combines FL and mimic learning to implement IDS. The main feature of this technique is that it minimizes the probability that the attacker can obtain sensitive information against reverse engineering attacks on the student model. In mimic learning, a student model is trained with a public dataset labeled with the teacher model, which is trained on the client's sensitive data. The authors used the NSL-KDD dataset to benchmark the performance of federated mimic learning with other techniques similar to those in the literature. The detection accuracy was 98.11% for the proposed algorithms.

However, in [79], the authors propose an IDS mechanism that combines Generative Adversarial Networks (GAN) and FL for smart home IoT systems. GAN generates synthetic data to augment the data on an IoT device that trains the IDS model on each device individually. Thus, GAN solves the problem of limited, missing, and unbalanced data in IoT devices. Using FL, a PPFL framework is constructed, enabling numerous IoT devices to participate in creating a thoroughly trained global IDS model. Each device introduces its single model using its own data and synthetic local data generated by the local GAN, and the model parameters are transferred to the global model. The data generated for each IoT device through GAN and the local data are not shared between the devices, and this step guarantees the privacy of the data and resources. The global model performs the aggregation and communicates the updated model with the IoT devices. The authors also present how to develop a robust, comprehensive, multiclass, binary IDS model with many communication rounds with IoT devices in a network. The authors tested the precision, accuracy, and convergence of their proposed model for IDS on three standard datasets: NSL-KDD, KDD-CUPPP99, and UNSW-NB15.

In [80], the authors propose an FL-based IDS for agricultural IoT called FELIDS. To implement FEILDS, the authors use three DL classifiers, namely Deep Neural Networks (DNN), Convolution Neural Networks (CNN), and Recurrent Neural Networks (RNN). Three real-world datasets, namely CSE-CIC-IDS2018, MQTTset, and InSDN, were used as benchmarks to evaluate the performance of each DL classifier to implement the IDS. The data sets mentioned here consist of large amounts of network traffic, both malicious and benign, and training on them helps to further develop the threat model. The authors also conducted a comparative performance analysis between FELIDS, centralized ML models, and other state-of-the-art studies.

The authors of [50] propose an efficient FL-based NIDS design, and the main contributions are as follows. (i) They propose an efficient model called dynamic weighted aggregation federated learning (DAFL) based on FL and use all features of FL for the privacy-preservation characteristics. (ii) DAFL differs from other FL-based IDSs as it implements

dynamic filtering and weighting processes for local models. (iii) DAFL is also efficient as it better detects network intrusions with less communication overhead.

In [95], the authors examine the distinctions between the challenge of NIDS and other areas where ML methodologies have enjoyed greater efficacy. The authors in [95] assert that the identification of attacks differs fundamentally from these applications, which poses significant hurdles to effectively integrate ML within the intrusion detection framework. The authors substantiate this argument by delineating the unique challenges inherent in network intrusion detection and offering recommendations to fortify future research efforts in anomaly detection.

As discussed above, many researchers have tried and implemented their versions of the state-of-the-art IDS for IoT systems using FL, DL, or traditional ML algorithms, intending to make the IDS for IoT systems more efficient and accurate.

V. PRIVACY PRESERVING LEARNING

This section provides a comprehensive analysis of the current literature on learning techniques with respect to the preservation of model and user data in context, representing the use of ML, DL, or FL techniques alone or in combination with any of the privacy preservation mechanisms discussed in Section II. The topics include non-FL-based privacy-preserving learning, PPFL, PPFL for IDS, and PPFL for IoT.

A. NON-FL BASED PRIVACY-PRESERVING LEARNING

This subsection discusses and summarizes some notable research papers in the literature on the category of non-FL-based privacy-preserving learning. For example, in [12], the authors have proposed a novel approach to detect attacks in IoT networks by employing a combination of 2-convolution neural networks (CNN) for feature selection and robust detection modeling in a centralized learning system without FL. In [41], the authors propose a lightweight and hybrid DL-based privacy-preserving model for the Industrial Internet of Medical Things (IIoMT) based on fog computing. IIoMT devices collect, monitor, exchange, and analyze the required data using open channels that are not properly encrypted. The authors in [41], [44] propose a two-pronged approach to resolve this problem. First, a blockchain-based scheme is designed in which all participating devices are registered, verified, and then validated using smart contract-based enhanced proof of work to achieve the goal of security and privacy. The second step is the design of a DL scheme with a Variational AutoEncoder (VAE) technique for privacy and Bidirectional Long Short-Term Memory (Bi-LSTM) for intrusion detection. The authors then tested their approach and implementation on available state-of-the-art datasets, and the performance was as expected. In the study [43], the authors present a pragmatic privacy preservation mechanism to safeguard data integrity in various phases of a wearable Internet of Things (IoT) framework, specifically focusing on the use of smart wearable devices. Initially, the authors used the lightweight PRESENT algorithm to encrypt information within IoT devices while leveraging pallier HE for data management on the cloud platform. Subsequently, during the data publication phase, the authors enhance the traditional k-anonymity algorithms for static data and incorporate the temporal-differential privacy algorithm for real-time data, followed by a rational exposition of the synergistic integration of these optimized algorithms.

In [60], the authors propose a prototype system focused on privacy preservation and data analytics that uses deep learning for health-related data analysis. It introduces a non-privacy data extraction algorithm to analyze health data post-privacy information separation and employs data augmentation techniques to prevent overfitting. In addition, a customized CNN is utilized to construct a security module.

In [71], the authors propose an efficient and privacy-preserving scheme for cloud security using enhanced cryptographic honeypots. The authors in [71] used the Gray-Level Co-occurrence Matrix (GLCM) for feature extraction, fuzzy K-means for clustering, CNN as a classifier, and cryptographic techniques using honeypots as privacy-preserving measures.

The authors in [93] present a framework for protecting privacy in smart contracts using blockchain and artificial intelligence (PPSC-BCAI) to streamline human interaction, system activities, service alerts, security risks, and fraudulent claims. The framework employs Extreme Gradient Boosting (XGBoost) to analyze data transactions and sharing, thereby optimizing network load by evaluating transaction services and discerning fluctuations in transmission rates across unreliable network connections.

The authors in [106] introduce an efficient privacy-preserving model designed to protect data storage in consumer edge devices, focusing on wearable sensor medical devices. The proposed model [106] utilizes a modified artificial neural network based on gray wolf optimization (GWO-MANN) to detect anomalies within the data. Subsequently, data encryption is performed using a novel dynamic attribute-based re-encryption (DABRE) model before storage in a cloud database, ensuring irretrievability and bolstering secondary data protection. The efficacy of the anomaly detection method is evaluated using metrics that include detection rate (DR), false alarm rate (FAR), and accuracy, with comparisons made with an existing artificial neural network (ANN) method.

One of the best examples of a non-FL-based application of privacy-preserving learning can be found in [129]. In [129], the authors discuss PPML in a scenario where PPML involves secure outsourced computing (SOC) due to high computational overload. In the SOC methodology, the computational tasks are outsourced to a set of powerful and specialized servers, which compute and evaluate the functions on a pay-per-use basis. To ensure that server transactions are secure and private, the authors of [129] propose SWIFT (Super-Fast and Robust PPML), a framework for a variety of ML

algorithms in the SOC setting, which ensures that users receive their output communicated to them irrespective of any adversarial behaviors encountered. Robustness, which the SWIFT framework offers in the 3-party computation (3)-PC) setting over the rings, ensures that the clients are guaranteed output delivery in a typical honest majority setting. SWIFT is considered the first robust and efficient PPML framework with 3-PC. In the paper, the authors show that SWIFT is as fast as the *fair* 4PC framework. The authors demonstrate the practical relevance of the SWIFT framework by benchmarking popular ML algorithms, such as Logistic Regression, and DNN, such as VGG16 and LeNet, both over a 64-bit ring in the WAN setting. For the DNN setting, the results verify the authors' claim that they provide an improved security guarantee while incurring no additional overhead for the 3PC and obtaining at least twice the improvement for the 4PC [129].

In the Non-FL-based privacy-preserving learning paradigm, one of the first papers published in 2015 on the topic of deep learning that preserves privacy (PPDL) was published [29]. This paper discusses the design, implementation, and evaluation of a PPDL framework and system based on stochastic gradient descent (SGD). The authors in [29] implement and design a practical system that enables multiple parties to jointly learn an accurate neural network model for a given problem without sharing the required input datasets. The authors are exploiting the fact that all the optimization algorithms used in modern deep learning, especially those based on SGD, can be parallelized and executed asynchronously. The system's unique feature is that it lets the clients train independently on their datasets and selectively share small subsets of the learning models' key parameters during their training. The proposed framework offers an attractive point in the privacy and efficiency trade-off debate. The clients preserve their privacy and that of their respective data while still profiting from other clients' models, boosting their learning accuracy beyond what is achievable only using their data inputs. The key technical innovation of the article is the selective sharing of the model parameters of the clients during training. The authors show the accuracy of their PPDL framework using various benchmark datasets [29].

The authors in [180] propose a new personalized 3D location privacy protection mechanism (P3DLPPM) to address the limitations of existing 2D location privacy protection techniques in the 3D space. The proposed mechanism operates in two phases: The first phase employs a 3D Hilbert curve-based minimum distance searching algorithm to formulate a Protection Location Set (PLS) to hide the actual site while guaranteeing DP. In the second phase, a novel Permute-and-Flip mechanism is introduced for location perturbation, which generates fake locations with smaller perturbation distances, thereby improving the trade-off between privacy and Quality of Service (QoS). The authors formally study the relationship between geoindistinguishability and distortion privacy (expected inference error) in the 3D space and leverage this relationship in designing the P3DLPPM. The simulation results demonstrate the ability of the mechanism to significantly improve personalized privacy protection while meeting the QoS requirements of clients [180].

The study in [181] proposes a novel mechanism that realizes 3D location privacy protection (LPP) in mobile crowd-sensing (MCS) applications, leveraging the notion of 3D geo-indistinguishability (3D-GI). This novel technique is designed for local implementation by the client, thereby obviating the requirement for a trusted third-party intermediary. It strives to achieve a balance between preserving location privacy and maximizing server profit, ultimately optimizing the utility of the Multi-Cloud System (MCS). The authors propose an Asynchronous Actor-Critic Mechanism (A3CM) that dynamically determines the optimal perturbation policy through an iterative trial-and-error process. This eliminates the need for prior knowledge regarding the system itself and potential attack models. The A3CM explores intricate 3D MCS environments effectively by identifying the optimal policy in the continuous policy space and employing an asynchronous multithread training model. Extensive simulations are conducted under two task allocation scenarios, involving single and multiple task completions per user, to validate the robustness of the mechanism and demonstrate its superior performance against benchmarks in terms of LPP and MCS system utility [181].

Table 4 lists the important non-FL privacy-preserving methodologies referenced in the literature. Concerning Table 4, in [104], the authors use a multiplicative depth-based HE approach, which is computationally expensive. In [101], the authors use IoT-ToN and BoT-IoT data sets to evaluate their scheme. In [102], a mathematical technique named PICO, that is, ''Privacy-Preserving Access Control in IoT Scenarios through Incomplete Information,'' is used as a non-FL-based privacy-preserving technique. In [105], the authors tested a two-level privacy scheme and it was validated on the IoT-ToN and BoT-IoT datasets. In [110], the authors implement an IDS mechanism for an SCADA data device while maintaining the privacy of the data. The authors in [111] use the Advanced Encryption Standard (AES) algorithm to secure the parameters and the SHA-256 algorithm for verification. The authors apply an object-oriented privacy guarantee, a pipeline-oriented privacy guarantee, and a non-FL-based privacy-preserving mechanism in [125]. In [118], the authors used the MedNIST dataset to evaluate their scheme. In [119], the authors have implemented a blockchain-based and smart contract-based electronic health record (EHR) system. The authors in [126] implement the SPEED framework that includes data distribution, privacy preservation, and redundancy minimization. In [131], the authors focus on trustworthiness, fairness, privacy and when their framework is concerned with multi-party transactions. The authors in [132] implement electronic proof of work and reputation-based trustworthiness mechanisms and use BoT-IoT and ToN-IoT data sets for evaluation. The authors in [180], [181] use reinforcement learning (RL) with A3C and

TABLE 4. Comparisons of non-FL privacy-preserving methodologies.

References	Non-FL Privacy-Preserving Framework	Learning Algorithms	PPM	Privacy Guarantee
[43]	k-Anonymity	PRESENT	Temporal DP, HE	Medium
[55]	Blockchain-DP framework	None	Blockchain, Local-DP	High
[62]	DP with DL	DP-SGD, DP-PCA	DP	High
[67]	DL-IDS-IoT	Fully Connected DNN	Filtered Layers	Good
[71]	Enhanced-Honeypot-Crypto	GLCM + K-Means + CNN	Encrypted Honeypots	Good
[93]	PBSC-BCAI	XG-Boost	Blockchain	Good
[101]	P2SF-IoV	LSTM	Blockchain	High
[102]	PICO	None	Attribute Based Access Control, AES, RSA, ABE	High
[104]	SoK	RNN, CNN	HE	Medium
[105]	PPSF	PCA, Gradient Boost	Blockchain-IPFS integration	High
[106]	DABRE	GWO-MANN	Encryption	Medium
[110]	PPID	EM Clustering	Co-relation Coefficient Technique	Good
[111]	Collaborative DL Architecture	SGD, DSGD, CNN	DP	Medium
[125]	Generic PPML Architecture	DP-SGD, NN-EMD, SA-FL, Hybrid Algos.	DP, HE, SMPC, Anonymization, Garbled Circuit, Additive Mask	Good
[118]	Blockchain-Based Distributed FL Architecture	SGD, CNN	Blockchain, SMPC, DP	Good
[119]	Distributed Healthcare Blockchain System	No Particular Algos.	Blockchain Smart Contracts	Medium
[126]	SPEED for IoV	LS-SVMs, CNN	SPEED + LS-SVM + PPDL	Good
[131]	DPPDL	GAN, DPGAN	DP + Blockchain	Good
[132]	TP2SF	PCA, XGBoost	Blockchain	Good
[180]	P3DLPPM	LPP + A3C algo. + RL	DP	Good
[181]	3D-LPP in MCS on 3D-GI	A3C algo. + RL + 3C based LPP	DP	Good

LPP along with DP to implement location-based 3-D privacy of clients in a mobile crowd-sensing scenario.

B. PRIVACY-PRESERVING FL

In this section, the authors discuss the state-of-the-art research on using PPFL in various scenarios and applications. They also discuss only some of the important studies in the field.

The authors in [46] present a distributed infrastructure to promote peer-to-peer trust among stakeholders, facilitating collaborative execution of privacy-preserving workflows. The authors in [46] propose a prototype that establishes industry gatekeepers and regulatory bodies as credential issuers within a specified application domain. Before engaging in distributed learning workflows, potential malicious actors must obtain valid credentials from these gatekeepers. A proof of concept that leverages Hyperledger Aries, Decentralized Identifiers (DIDs), and Verifiable Credentials (VCs) to construct a distributed trust framework during a privacy-preserving machine learning endeavor. The methodology relies on secure and authenticated DID communication channels to facilitate a federated learning process pertinent to mental health data.

In [56], the authors discuss the modest success achieved so far in the application of ML and blockchain technology in medicine, with a shift in focus towards exploring its intersection alongside privacy-preserving techniques. It emphasizes the potential of combining these technologies with FL to advance evidence-based medicine without compromising the privacy of highly regulated personal health information. The authors in [56] outline a brief history of these medical technologies, identify challenges, and propose a framework for an advanced blockchain-orchestrated ML system for PPFL in medicine. Six critical elements for this approach are highlighted, including ensuring that data and analytic processes are discoverable on secure public blockchains while maintaining privacy, incentivizing data quality through tokenized reputation-based rewards, and preventing model poisoning attacks.

In [57], the authors highlight existing privacy-preserving methods, often centralized but decentralized, which pose security vulnerabilities. The authors in [57] introduce ModelChain, a novel framework utilizing blockchain technology that enables privacy-preserving machine learning by facilitating collaborative model parameter estimation across institutions without disclosing patient health information. Integrating privacy-preserving online machine learning with a private blockchain network enhances interoperability. It supports national healthcare priorities such as Patient-Centered Outcomes Research (PCOR), addressing both the benefits and potential issues of this approach.

In [63], the authors present an FL framework designed to facilitate the extraction of a global model from healthcare data distributed in various localities. This framework in [63] provides dual layers of privacy protection. Primarily, it refrains from relocating or disseminating raw data amongst sites or to a centralized server during the model training phase. In addition, it employs a DP mechanism to further protect the model against potential privacy breaches. Through a comprehensive evaluation conducted on two healthcare applications, using authentic electronic health data sourced from a cohort of 1 million patients, the authors substantiate the feasibility and efficacy of the FL framework to provide an enhanced level of privacy while preserving the utility of the global model.

In light of the expanding landscape of edge computing applications in the realms of 5G and 6G, where ensuring privacy protection is not just desirable but imperative, the authors of [64] have conducted a comprehensive review that covers the fundamental underpinnings, central developmental problems, and existing DP-based methodologies relevant to edge FL. These discussions naturally align with the envisaged DP-Enhanced Federated Learning (DP-EFL) framework upon which our focus is directed. More specifically, in [64], the authors divert the readers' attention toward mitigating

noise within the DP-EFL paradigm, leading to the classification of existing efforts into 3-distinct categories. Furthermore, employing insightful analyses and deliberations concerning the impact of noise magnitude on the learning efficacy of DP-EFL systems, the authors have delineated challenges, identified promising avenues for resolution, and outlined avenues for future research aimed at advancing the field of DP-EFL.

Ensuring privacy in vertical-FL poses formidable challenges owing to the communication of intermediate outputs and model update gradients, rendering the underlying data of participating entities vulnerable to inference by adversarial entities. Consequently, the focus of the study in [65] is to elucidate mechanisms to safeguard the privacy of individual organizational data within a DP framework. Through experimentation conducted across various real-world datasets and DP budget configurations, the authors' findings in [65] underscore the necessity of striking a delicate balance between vertical-FL performance and privacy preservation, particularly concerning the level of perturbation noise introduced.

In [97], the authors proposed an efficient and robust privacy-preserving framework for cross-device FL based on the *tMK-CKKS* HE scheme. This scheme protects users' privacy by ensuring that no users can reveal their model parameters during FL secure aggregation. The proposed framework is robust for users who join and drop out. The security analysis shows that it is secure in an honest but careful setting, even under collusion and collaboration attacks between up to $t - 1$ users and the centralized server, where t refers to the number of users. The evaluation was done on image classification and text prediction, conducted on realistically partitioned datasets to demonstrate that the proposed scheme preserves model accuracy better than traditional FL and is more efficient as far as computation and communication go as compared with existing HE-based PPFL. However, the proposed framework will not be able to protect the model's privacy after training, which is the central theme of DP.

In [113], the authors propose a PPFL scheme based on CKKS-HE. They take the standard Fed-Avg algorithms and present a privacy-enhanced version by infusing the CKKS-HE scheme for the privacy and security of model parameters. The scheme achieves the same training effect as Federated Averaging, but takes some extra time. Still, it performs better with respect to communication and computation cost than the Paillier-based HE Fed-Avg. Many experiments were conducted with two real-life data sets to evaluate the scheme's effectiveness, and the advantages of communications and computation overheads were shown.

In [122], the authors propose a three-plan framework to solve the problem of user-level privacy leaks under the cross-silo FL framework to secure the cross-silo FL by taking advantage of the Local Differential Privacy (LDP) technique. The main insight is that LDP can protect data privacy while retaining client data statistics to preserve its high quality and utility. The framework can be deployed as an auxiliary module to adapt to existing FL systems to improve privacy. The results of the experimental evaluation validate and prove the effectiveness of the proposed framework.

In [124], the authors want to find an appropriate solution to the problem of sharing data on safe and secure network traffic analysis between various collaborators for analysis. The paper provides a practical study on how HE and SMPC can be combined to share network traffic analysis data securely and privately among various clients for network traffic analysis and prediction. First, the FL model aggregation process is combined with the CKKS-HE scheme. Then, SMPC is implemented to securely share the network traffic analysis with various users and clients involved for evaluation. The results and analysis show that the loss of precision does not necessarily reduce the accuracy every time; it can even increase in some specific cases. Encryption can cause a slight computation overhead, but it can be an open research question for other researchers.

In [135], the authors propose a new quality-dependent aggregation procedure to eliminate bad quality updates. The method allows users to exchange their model updates directly without involving a central server and autonomously aggregate the received model updates without needing a global sync procedure. In addition, the method can effectively detect whether an update received by a user is of poor quality. The authors introduce two new techniques: a quality-based aggregation method and an extended dynamic [50] and [61] used CNN as the ML technique but used dynamic weighted averaging FL and DeepFed as their frameworks and PPMs. Adding to the 2-methods and LDP, the authors propose a reliable and privacy-preserving decentralized FL scheme designed to support batch joining or leaving of clients while incurring minimal delay and achieving high model accuracy and precision. The results of the experimental simulation demonstrate that the proposed scheme ensures an optimal trade-off between the accuracy of the model and the privacy of the data.

C. PRIVACY-PRESERVING FL FOR IDS

This subsection discusses recent research on implementing PPFL to apply IDS in various scenarios. Table 5 gives an overview of the state-of-the-art literature referenced and surveyed in this paper covering *"the applications of PPFL for implementing IDS in various scenarios"*. In this section, we also summarize the literature in Table 5 and discuss some crude details about the surveyed literature. Taking into account the entries in Table 5, we discuss some important works in the literature related to PPFL-IDS. In [47], the authors introduced a framework that uses adaptive VFL using DP combined with support vector machine, kNN, MLP, and random forest ML algorithms and techniques. References [50] and [61] used CNN as the ML technique but used dynamic weighted averaging FL and DeepFed as their frameworks and PPMs. The authors in [68] use encoding and FL as a PPM, while the authors in [74] use FL plus

TABLE 5. Comparison of state-of-the-art literature on PPFL for IDS.

References	Framework	Learning Algorithms or Techniques	PPM	Privacy Guarantee
[47]	Adaptive VFL using DP	SVM, MLP, kNN, RF	DP	Good
[50]	Dynamic Weighted Averaging FL	CNN	Weighted Parameter Averaging using CNN + FL	Good
[61]	DeepFed	CNN	Weighted Parameter Averaging using CNN+FL	Good
[74]	F-BIDS	DT, RF, CNN	FL + Model Blending	Fair
[76]	FL based Mimic Learning	FTML, FSML, FF-MLP-NN	FL Based Mimic Learning	Good
[78]	F-AL-Based-IoV-IDS	ConvLSTM + DRL based FL client selection	Only FL itself	Good
[79]	FED-GAN-IDS	GANs + FL	FL+GAN itself	Good
[80]	FEILDS	DNN + RNN + CNN	Trustworthy FL Aggregation Algorithms	Good
[11]	FL-IIoT-IDS	CNN-RNN	FL-Aggregation	Good
[82]	FLDID-IIoT-IDS	CNN + LSTM + MLP	HE with Paillier based Encryption for Communications	Good
[84]	GowFed IDS	Gower-Distance Matrix Learning + MLP	Gow Matrices + FL	Good
[86]	HFL based IDS for IoT	SGD + NN	FL + HFL	Good
[89]	FL based IDS for IoT	Simple ML Algorithms for Prediction and Classification	Just FL	Fair
[90]	FL based IDS for IIoT	Simple ML Algorithms for clients and Aggregation Server	FL + DP	Good
[98]	MITFeD-using multiple-SDN domains for IDS	SVM + FL Aggregation	SMPC + Blockchain	Good
[100]	P2IDF IDS for SDIoT-Fog	ANN + SAE	SAE + ANN + SDN + FL	Good
[103]	Poisoning attack on FL based anomaly detection	Simple ML Algorithms + FL	DP + Generalized Clustering Approach + Filtering data	Fair
[109]	Secure FL for Network Traffic Prediction	Simple ML Algorithms + FL	HE using CKKS + SMPC	Good
[134]	DP-FED-AVG-GAN + Autoencoders	GAN	DP + GAN	Good
[137]	ShieldFL Mitigating Model Poisoning	Only FL aggregation using simple ML algorithms	2-trapdoor HE + Byzantine-Tolerance Aggregation Mechanism	Good
[138]	DPA-FL-Poisoning Attack Defence	LOF + RL + FL	Just FL	Fair

model blending as a PPM. The authors in [80] present a novel framework, FEILDS, with trustworthy FL-based aggregation algorithms as a PPM. In [76], the authors use FL-based mimic learning, a unique concept as a PPM, to implement FL-based IDS. In somewhat novel scenarios, the authors in [79] and [134] use Generative Adversarial Networks (GANs) with FL and DP, respectively, as a PPM. While [11], [78], [89], and [138] only use FL as a PPM to implement IDS in their respective frameworks, namely F-AL-Based-IoV-IDS, IDS-for-IIoT, IDS-for-IoT, and DPA-FL-Poisoning-Attack-Defense. Different schemes and flavors of HE, namely, HE with Paillier-Based Encryption, HE with CKKS plus SMPC, and 2-trapdoor HE with Byzantine tolerance-aggregation mechanism, the authors used in [82], [109] and [137] as PPM, respectively, to implement their respective FL-based IDS. Here, we mention some of the critical research articles in Table 5, with some unique implementation characteristics and properties; however, other researchers cited in Table 5 also had some impressive approaches to implement IDS using PPFL.

D. PRIVACY-PRESERVING FL FOR IOT

This subsection discusses state-of-the-art research work on implementing PPFL for various avatars and use cases of IoT. Table 6 gives the audience an overview of the state-of-the-art literature referenced and surveyed in this paper covering "*the applications of PPFL for IoT systems*." In this section, we also summarize the literature in Table 6 and discuss some crude details about the literature surveyed. Considering the entries in Table 6, we try to explore how researchers have implemented various PPFL mechanisms in different kinds of IoT systems. In [40], [44], and [45], the authors used simple ML-Algorithms for on-device training and central aggregator training but have different frameworks and PPMs to implement PPFL for IoT systems to improve the privacy and efficiency of the IoT systems. In [68], the authors proposed and implemented an FL-based framework named FEDMAL, which used an efficient and lightweight CNN named AM-NET combined with FL for malware detection in IoT systems. The authors of [49], [51], and [108] use FL as the sole PPM to preserve the privacy of the IoT systems. References [107] and [115] use Generative Adversarial Networks (GANs) along with FL and Blockchain, respectively, as PPMs to preserve the privacy of their particular IoT systems. The authors in [115], [130], and [140] use Blockchain plus FL for the first two and Blockchain plus FL plus DP as PPMs to preserve the privacy of their particular IoT systems. In [178], the authors propose several local differential privacy (LDP) mechanisms: PM-OPT (an optimal piecewise mechanism), PM-SUB (a sub-optimal mechanism), Three-Outputs, and HM-TP. These mechanisms aim to preserve privacy when collecting data records and computing accurate statistics for various data analysis tasks, including estimating mean frequency, support vector machine (SVM) classification, logistic regression, and linear regression. The proposed LDP mechanisms are integrated with the FedSGD algorithm, creating an LDP-FedSGD algorithm for vehicular crowdsourcing applications to train machine learning models for predicting traffic status while mitigating privacy risks and reducing communication costs. By leveraging LDP mechanisms, adversaries are prevented from deducing the exact location information of vehicles from the uploaded gradients. Furthermore, FL enables vehicles to train local machine learning models using collected data and transmit noisy gradients, instead of raw data, to the cloud server to obtain a global model. Extensive experiments demonstrate the effectiveness and superior performance of the proposed approaches compared to existing solutions. Future work aims to apply the proposed LDP mechanisms to deep neural networks for more complex data analysis tasks. In [179], the authors put forward an innovative framework to facilitate the training of LDP clustered FL models on heterogeneous IoT data. To mitigate the deleterious impact of LDP noise, the authors in [179] proposed an adaptive layer-wise weight perturbation approach, which injects LDP noise at a finer granularity by perturbing each weight individually, thereby enhancing the model's utility. To circumvent the pernicious

TABLE 6. Comparison of state-of-the-art literature on PPFL for IoT.

References	Framework	Learning Algorithms or Techniques	PPM	Privacy Guarantee
[40]	PPFL Architecture for IoT Health-Care System	Basic ML Algorithms for On-Device and Central Aggregator Training	Blockchain + FL	Good
[44]	BPFL for PPFL in IoV	Basic ML Algorithms for On Device and Central Aggregator Training	Multi-Krum + Blockchain + FL + Piallier Crypto-system	Good
[45]	Trustworthy PPFL for IIoT	Basic ML Algorithms for On-Device and Central Aggregator Training	FedML + DP + Smart Contracts + IPFS + Blockchain	Good
[49]	Ensemble Deep FL Model	Ensemble Deep Learning for FL	Just FL	Good
[51]	MV-FLID-IoT-IDS	GWO + ANN Classifiers + RF	Just FL	Fair
[68]	FED-MAL	AM-NET-CNN + FL	Just FL	Good
[88]	HE Based PPFL for IoT Healthcare	DNN+CNN+RNN	HE+FL	Good
[107]	PPFLDL-for-Fog-IoT	GAN	FL + GAN	Good
[108]	PPFDL-for-Wearable-IoT-Biomedical	SVM+MLP+FFT+FL	FedAvg for FL	Good
[115]	PPFL-IEoT-Attack-Detection	TC-GAN	Blockchain-Based FedAvg	Good
[130]	Secure PPFL for IoT	Basic ML Algorithms for On-Device and Central Aggregator Training	Blockchain Integrated with FL	Good
[140]	DP-Based-Asynchronous-PPFL	Basic ML Algorithms for On-Device and Central Aggregator Training + PBFT + FL	DP+FL+Blockchain	Good
[178]	LDP Based FL-SGD	LDP + FL-SGD + PM-OTP + PM-SUB	LDP	Good
[179]	ACS-FL with LDP	Clustered FL + LDP + DCT-Compression	LDP	Good

curse of privacy budget explosion, a parameter shuffling mechanism is employed that enables each client to shuffle the weights of their local model and transmit each weight through an anonymous channel to the server. Furthermore, we employed discrete cosine transform-based compression to compress the LDP model, thereby reducing the noise magnitude and communication overhead. The empirical evaluations in [179] demonstrate that the proposed ACS-FL (adaptive clipping, weight compression, and parameter shuffling) scheme achieves superior performance in striking an equilibrium between privacy preservation and utility maximization. As discussed above, many other referred literature in this survey paper use various PPMs and different frameworks to preserve privacy and improve the efficiency of their implemented PPFL-based IoT systems.

VI. CASE STUDIES ON APPLICATION OF PPFL-IDS-IOT

This section highlights some important case studies where one can implement PPFL for IDS in various IoT systems. Figure 12 illustrates how PPFL is used for various case studies discussed in this section.

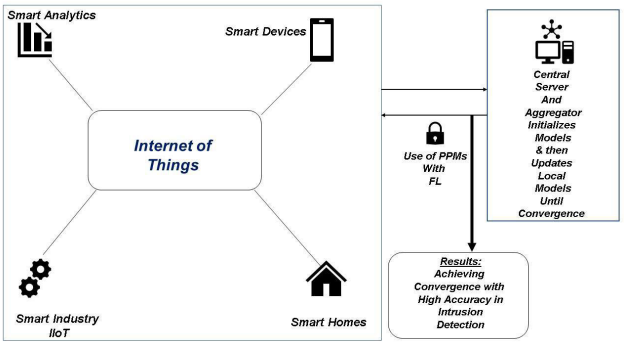


FIGURE 12. Example case-study graphical depiction for PPFL-IDS-IoT.

1) *Security and Privacy for Smart-Home-IoT Systems*: A smart home security system uses a variety of heterogeneous IoT devices, such as security cameras, motion sensors, and smart locks, to monitor and protect a client's home. Implementing a PPFL approach allows the system to collect data from multiple homes, analyze it collectively to identify potential threats, and develop a comprehensive understanding of common attack patterns across IoT networks. By doing so, the system can improve its ability to detect and prevent cyber-attacks while maintaining the privacy of individual users. For example, in [91], the authors proposed a smart-home architecture called IOTFLA. The authors use FL and secure data aggregation to implement this architecture, which consists of a hub, an IDS, a database, and a router. IOTFLA gives users a better hold and control over their smart-home network and devices and uses FL to improve the system's overall efficiency over time in a privacy-preserving manner [91]. Thus, PPFL can be used to improve the privacy and efficiency of any smart-home system [186].
2) *Security and Privacy for Medical-IoT Devices*: Medical-IoT devices connected to the Internet are vulnerable to cyber-attacks that can compromise patient safety and data privacy. By employing a PPFL approach, healthcare providers can aggregate data from multiple medical IoT devices to identify potential threats and develop a more effective IDS. This approach would empower healthcare providers to improve the security of their medical IoT device networks while maintaining the privacy of patient data and the integrity of medical IoT devices. We provide examples from the literature referenced in this survey paper to support this case study. In [77], the authors proposed and implemented an FL-based-blockchain-enabled task scheduling (FL-BETS) framework while considering different and constantly changing heuristics. The authors in [54], a blockchain-based approach for healthcare data sharing, emphasize user control over data access. It describes organizing network nodes into clusters for improved scalability and throughput, with each user assigned to a specific cluster. A hierarchical system based on user-assigned IDs is also introduced to streamline search and access to relevant data clusters. The authors in [77] considered different health-based applications with hard and energy-resource consumption constraints during execution and production on the distributed fog and cloud nodes. The main goal of FL-BETS in [77] is to identify and ensure the privacy preservation and fraud-detection in the data at various levels, for example, at local fog nodes and remote cloud nodes, using minimum energy consumption and delay, and to

satisfy the deadlines of healthcare workloads. Another example of the application of the social - internet of medical things is in [136]. In [136], the authors proposed a lightweight-2-factor authentication framework (L2FAK) with privacy-preserving functionality that uses mobile-sink for smart-eHealth systems. The authors have provided formal and informal analyses of the proposed L2FAK to resist cyberattacks such as session stealing, message modification, and denial of service, protecting the devices and data integrity. Learning analysis on L2FAK, in [136], verifies physical layer features using FL-layered authentication. The implementation results show that FLLA (Federated-Learning Layered Authentication) and L2FAK analyze the protection of privacy features effectively and achieve better quality metrics, like transmission efficiency and overhead ratio, than other state-of-the-art approaches.

3) *Security and Privacy for IIoT Devices*: Industrial control systems (ICS), supervisory control and data acquisition (SCADA) devices, and other IIoT devices are used in many industries, such as manufacturing, energy, transportation, and critical sectors crucial to the normal functioning of a city or even a country. ICS and SCADA networks mainly consist of various IIoT devices like industrial sensors, PLCs, and infrared and x-ray cameras on multiple machines. The ICS networks and interconnected IIoT devices are primarily vulnerable to cyber-attacks from adversaries like competing organizations and hostile nation-states. Any cyber-attack on ICS, SCADA, and networked IIoT can result in operational downtime, safety accidents, and perhaps financial losses for an organization. By employing PPFL, ICS operators can collect data from different systems and IIoT interconnected devices to identify potential threats and attack signatures and implement more efficient and capable IDS for ICS, SCADA, and IIoT devices. PPFL approach can enable operators to improve the security of their ICS and SCADA networks while maintaining the privacy of individual systems. From the referenced literature, we point out the following works from [11], [41], [45], [49], [52], [61], [90], [115], [165], and [172] to show the importance of PPFL in implementing IDS for different kinds of IIoT devices which enhances the privacy and security of IIoT devices at various levels.

VII. FUTURE RESEARCH DIRECTIONS AND OPEN RESEARCH QUESTIONS

In this section, we list and discuss future research directions and open research questions. In writing this section, we considered ongoing research directions and the need to develop PPFL for applications to implement IDS in IoT environments.

A. FUTURE RESEARCH DIRECTIONS

Here are some of the relevant future research directions that the research community must pursue in the PPFL domain for the implementation of IDS in IoT environments.

1) *Data Heterogeneity*: The data generated by various IoT devices may have different formats, structures, and properties, making it challenging to develop a unified intrusion detection model. Therefore, PPFL models should be designed and implemented to handle such data heterogeneity [186].
2) *Privacy and Security*: Privacy and security are paramount considerations in using PPFL for IDS implementation in IoT environments, which involve diverse devices collecting sensitive data, necessitating intrusion detection for threat identification. However, centralizing data for analysis poses substantial privacy risks, making privacy-preserving federated learning crucial. FL ensures sensitive data remains on devices, reducing breaches and unauthorized access. Additionally, federated learning empowers data owners with control over their data, mitigates data leakage risks, and accommodates the distributed nature of IoT. Adversarial threats like model inversion attacks are countered through robust privacy-preserving mechanisms, aligning with regulatory compliance such as GDPR. Secure model aggregation using cryptographic techniques ensures tamper-resistant transmission, fostering trust between participants. Scalability is achieved by distributing computation across devices and maintaining privacy while detecting intrusions effectively. Overall, privacy and security form the foundation for successful implementation in IoT environments, safeguarding sensitive data and facilitating collaborative intrusion detection.
3) *Model Aggregation*: The global model must aggregate the locally trained model parameters from the participating IoT devices in FL. During the aggregation, sensitive information may be leaked without proper security mechanisms. Therefore, secure model aggregation can be a very captivating area in the research domain. Although much work has already been done in this area, there is still a need for much more research to make secure model aggregation more efficient. In [67], [70], [105], [124], [125], [130], [145], [158], [159], [160], and [174], the authors use various cryptographic mechanisms, differential privacy, homomorphic encryption techniques, blockchain techniques and SMPC techniques to implement secure model aggregation.
4) *Communication Overhead*: FL involves frequent and overbearing communication between the client devices and the central aggregation server. This results in high communication overhead, which is challenging to handle in low-power and low-bandwidth IoT devices. Thus, this can be a very emerging future research emphasizing green communications, low-power communications, and computing in tune with the 2030 United Nations Sustainable Development Goals.

5) *Model Poisoning*: Adversaries may inject malicious data into the training dataset to compromise the federated learning model. The model must be robust against such attacks. Given that PPFL for IDS on IoT has in its base FL for its operation, detecting and defending against model poisoning attacks is an open research area to explore and add the solutions to the repository of the solutions to protect against the attacks.
6) *Model Interpretability*: Federated learning models lack interpretability, which makes it challenging to identify the cause of an intrusion. Thus, there is a need for the development of interpretable FL models that can explain the intrusion detection process in an organized way. Therefore, model interpretability is an open research problem.
7) *Local and Global Updates*: Federated learning models may have local updates (i.e., device-level updates) and global updates (i.e., central server updates). The optimal balance between local and global updates remains an open problem in federated learning.
8) *Real-time Detection*: IDS in IoT systems require real-time detection to prevent cyberattacks. However, due to communication and computation overhead, real-time detection may be challenging in FL systems. Therefore, developing real-time intrusion detection models in FL is still an open problem [185].
9) *Zero-Day Attack Detection*: Zero-day attacks refer to exploits and vulnerabilities that are unknown to the public or vendors and, therefore, have no patches or fixes. This makes the attacks extremely difficult and tricky to detect. As the IoT grows, detecting and preventing zero-day attacks on devices in their various avatars becomes increasingly essential. There are multiple ways or processes to detect the attacks. One of the ways is by using ML techniques. ML models can be trained to detect zero-day attacks by analyzing large amounts of data and identifying patterns illustrative of malicious activity. For example, an ML algorithm can be trained to detect when an IoT device is in use to launch a DDoS attack by analyzing and identifying traffic patterns consistent with a DDoS attack. Similarly, DL algorithms can be trained to detect abnormal or dishonest IoT devices. This can be extended to FL and, in a prominent manner, to PPFL, using enhanced privacy-preserving mechanisms like DP, HE, TEE, and Blockchain. The detection and prevention of zero-day attacks can be somewhat deciphered. Nonetheless, in the detection and prevention of zero-day attacks in IoT networks, especially IIoT networks where heterogeneous IoT devices are forming the network, no one knows with certainty whether all the devices from different manufacturers are safe and honest in all the training and learning scenarios. FL, with enhancements from other privacy-preserving mechanisms, as discussed in this paper, can help researchers design frameworks and techniques to detect zero-day attacks. Thus, zero-day attack detection in IoT environments is essential and a hard-open research problem [185].

B. OPEN RESEARCH QUESTIONS

Below are a few open research questions that emerged while conceiving and writing this survey. We suggest that researchers approach these questions and try to find new ways to solve them. We also recommend the listed open research questions based on the surveyed literature.

1) How can we design new PPFL algorithms for IoT-based IDS without compromising data owner's privacy?
Suggestions from Literature: Designing PPFL algorithms for IoT-based IDS without compromising data owner's privacy requires careful consideration of various techniques and principles. Based on this survey, we can answer this research question with the help of our survey and the privacy-preserving techniques mentioned in Section II. We can integrate differential privacy mechanisms to protect individual data during training. By adding controlled perturbations to the data, DP ensures that the contribution of any single user device to the federated learning process remains private [36], [37], [38]. We can also implement secure aggregation techniques to protect the aggregated model updates from individual devices. Utilizing novel cryptographic methods like HE or SMPC to aggregate model updates without exposing the details of each device's contribution. On the other hand, shifting more training tasks to the local devices reduces the amount of raw data transmitted. One can also have model updates only be shared by allowing local training, minimizing the risk of privacy breaches associated with sharing raw data [47], [48]. Researchers can use HE or secure channels to encrypt the communication between devices and the central server, ensuring that even the transmitted model updates remain confidential. One can also use TEEs to provide a secure environment for running code on devices, protecting the confidentiality and integrity of locally performed computations. One can also use devices with more relevant or diverse data that can contribute more significantly to the model, allowing for adaptive adjustments in the learning rates to ensure fair participation. Aggregating and summarizing data locally before sharing it with the central server helps preserve privacy by transmitting only essential information. One can also have devices that choose to participate in certain rounds of federated learning based on conditions, enabling greater control over when and how their data is used. Frequent model updates reduce the amount of information transmitted in each round, limiting the exposure of sensitive data. Researchers can design algorithms with privacy regulations and ethical considerations, ensuring the developed solutions comply with legal and ethical standards [132], [133]. The guidelines above provide a foundation for developing

PPFL algorithms for IoT-based IDS, focusing on preserving and protecting data owners' privacy. However, implementation details may vary based on specific use cases and system requirements.

2) How can we formulate new ways to evaluate the performance of the PPFL algorithm in terms of accuracy, efficiency, and privacy?
Suggestions from Literature: Evaluating the performance of PPFL algorithms requires a comprehensive approach that considers accuracy, efficiency, and privacy. As per this survey, in this paragraph, we discuss some methods to evaluate the performance of accuracy, efficiency, and privacy. Some of the performance metrics can be defined as follows: privacy metrics, efficiency metrics, convergence speed, robustness metrics, energy consumption, user experience, trade-off analysis, and real-world simulations. Within these performance metrics, we can define some new performance metrics as follows: differential privacy parameters, information leakage, membership inference attacks, global model performance, local model consistency, fairness metrics, communication overhead, computation time, resource utilization, number of rounds, convergence rate, adversarial training performance, model transferability, energy consumption per round, latency, model availability, privacy-accuracy trade-off, privacy-efficiency trade-off, simulation realism, and scalability [127], [173], [174].
3) How can we identify novel security and privacy threats to the PPFL algorithms and mitigate them?
Suggestions from Literature: PPFL is an evolving field that aims to ensure data privacy during the collaborative training of machine learning models. Identifying and mitigating security and privacy threats to PPFL algorithms is crucial to ensure the approach's effectiveness. Based on this survey paper, we suggest techniques for future researchers to identify and mitigate security and privacy threats to the PPFL algorithms. Some methods are discussed here but are not restricted to the following: adversarial testing and simulation, adding noise makes it harder for the attackers to attack; the use of HE, SMPC, and other encryption techniques can also help; the use of secure aggregation processes, use of continuous monitoring and update, maintaining transparency in the FL process, and allow for auditing of the system to ensure compliance with privacy and security standards of the system being operated upon, and continuous collaboration and research with the experts also helps [113], [114], [115], [143], [146].
4) How can we optimize the PPFL algorithms to reduce the communication overhead and improve the convergence rate?
Suggestions from Literature: Reducing the communication overhead and improving the convergence rate are critical aspects of optimizing PPFL algorithms. Here, we discuss some techniques one can deploy to achieve these goals based on this survey paper to help future researchers solve this research question. Model compression techniques, local model updates, asynchronous updates, smart selection of clients, gradient aggregation techniques, local training and fine-tuning, adaptive learning rates, heterogeneous federated learning, and some hybrid techniques like combining FL with traditional FL approaches can be some of the strategies researchers can apply to significantly optimize the PPFL algorithms, minimizing communication overhead and improving convergence rates without compromising privacy [117], [118], [150], [174].
5) How can we extend PPFL techniques to handle diverse and dynamic IoT environments with changing device types and numbers?
Suggestions from Literature: Extending PPFL techniques in varied, diverse, and dynamic IoT environments with changing device types and numbers requires a flexible and adaptive approach. We recommend to the researchers some approaches that can be used to extend PPFL techniques to handle diverse and dynamic IoT environments with variable device types and numbers. The approaches mentioned above and techniques are the following: the use of FL frameworks for heterogeneity, adaptive model design, dynamic participation, and device management; the use of edge computing and on-device learning, the use of transfer learning and model personalization, the use of privacy-preserving techniques for heterogeneous data, the use of de-centralized federated learning, the use of dynamic hyper-parameters and dynamic learning rates, use of energy-efficient communication, applying security measures when on-boarding a new device to the FL network and finally the use of continuous monitoring and adaptation techniques [88], [89], [90], [111], [112], [113], [133], [150], [151], [152], [153], [154], [155], [163], [164], [166], [169].

VIII. CONCLUSION

This survey paper studies state-of-the-art peer-reviewed research on federated learning (FL), privacy-preserving FL (PPFL), PPFL-IDS, and PPFL integration in IoT systems. The growing attack surface on critical infrastructure due to numerous, diverse IoT devices motivates the need for secure, privacy-preserving IDS solutions. We discuss various privacy-preserving mechanisms like homomorphic encryption (HE), differential privacy (DP), blockchain, trusted execution environments (TEEs), and secure multi-party computation (SMPC). We then delve into FL, its variants, and privacy-preserving nature, followed by threats and attacks specific to FL and PPFL. The survey covers the latest advancements in PPFL, PPFL-IDS, PPFL-IoT, and non-FL-based privacy-preserving learning. We present case studies demonstrating PPFL-based IDS implementations for IoT systems. Finally, we identify open research questions for

future exploration and provide suggestions based on the literature. The key challenge lies in detecting and preventing zero-day attacks due to IoT devices' heterogeneous and non-standard nature. Other critical areas include developing more efficient and scalable PPFL algorithms, designing robust and attack-resistant PPFL protocols, conducting comprehensive evaluations of PPFL-IDS systems, and fostering collaboration between researchers and industry to realize practical PPFL-IDS solutions. This survey aims to comprehensively understand PPFL and its potential for implementing IDS in IoT systems. We hope to motivate further research in this crucial domain and believe PPFL-based IDS systems hold immense promise. The ongoing expansion of IoT device deployment underscores the escalating importance of robust and privacy-conscious IDS solutions. We encourage the research community to address these challenges and look forward to witnessing future advancements in this field.

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ABHISHEK VYAS (Graduate Student Member, IEEE) received the Bachelor of Technology (B.Tech.) degree in computer science and engineering from Suresh Gyan Vihar University, Jaipur, Rajasthan, India, in 2016, and the Master of Technology (M.Tech.) degree in computer science and engineering with a specialization in information security from Vellore Institute of Technology, Vellore, Tamil Nadu, India, in 2018. He is currently pursuing the Ph.D. degree with the Department of Computer Science and Information Engineering, National Chung Cheng University, Taiwan. During his undergraduate studies, he also studied for some years as an international student at The Ohio State University, Columbus, OH, USA, and Florida Institute of Technology, Melbourne, FL, USA. He was an Ad Hoc Lecturer with the Department of Computer Science and Engineering, JK Lakshmipat University, Jaipur. His research interests include network security, systems security, next-generation networks, and applications of AI in cybersec and infosec domains. He is a Graduate Student Member of the IEEE Computer Society and IEEE ComSoc. He is a member of ACM. He is an Associate Member of the Institute of Engineers, India. He is a Certified Chartered Computer Engineer by the Institute of Engineers.



PO-CHING LIN (Member, IEEE) received the Ph.D. degree in computer science from National Chiao Tung University, Hsinchu, Taiwan, in 2008. He joined as a Faculty Member with the Department of Computer Science and Information Engineering, National Chung Cheng University, Taiwan, in August 2009, where he is currently a Professor. His research interests include network security, network traffic analysis, and network system performance evaluation. He also sits on the IEEE Communications Surveys and Tutorials Editorial Board.



REN-HUNG HWANG (Senior Member, IEEE) received the Ph.D. degree in computer science from the University of Massachusetts, Amherst, MA, USA, in 1993. He is currently the Dean of the College of Artificial Intelligence, National Yang Ming Chiao Tung University (NYCU), Taiwan. Before joining NYCU, he was with National Chung Cheng University, Taiwan, from 1993 to 2022. He has published more than 300 international journals and conference papers. His research interests include deep learning, AI adversarial attacks, network security, wireless communications, the Internet of Things, cloud, and edge computing. He is also on the IEEE COMMUNICATIONS SURVEYS AND TUTORIALS and IEEE TRANSACTIONS ON VEHICULAR TECHNOLOGY editorial boards. He received the Outstanding Technical Achievement Award of the IEEE Tainan Section in 2022 and the Outstanding Research Award of TACC, Taiwan, in 2023. He received the Best Paper Award from the Sixth International Conference on Internet of Vehicles 2019, IEEE Ubi-Media 2018, IEEE SC2 2017, and IEEE IUCC 2014, and the IEEE Outstanding Paper Award from IEEE IC/ATC/ICA3PP 2012.



MEENAKSHI TRIPATHI (Senior Member, IEEE) received the Ph.D. degree in computer science and engineering from the Malaviya National Institute of Technology Jaipur, Jaipur, Rajasthan, India, in 2015. She is currently an Associate Professor with the Department of Computer Science and Engineering, Malaviya National Institute of Technology Jaipur. She has more than 15 years of teaching and research experience in computer science and information security. She has published more than 100 research papers in leading journals, conference proceedings, and books, including IEEE TRANSACTIONS, *International Journal of Communication Systems*, *Computer Networks*, and *The Journal of Supercomputing*. Under her guidance, four students have already been awarded their Ph.D. degrees, and five are currently working with her. She has supervised around 33 PG students and over 60 UG students for their project work. She holds several professional designations, including the Dean of Rajasthan ILD Skills University, Jaipur, Rajasthan, India; a BoS Member with the University of Kota, Rajasthan; the Ex-Chairperson of the Computer Society of India (CSI), Jaipur Chapter, and a BoS Member of the Central University of Rajasthan, India. Her research interests include information security, wireless sensor networks, the IoT, SDN, and blockchains. She is a Senior Member of ACM and a Lifelong Member of CSI. She has organized various faculty development programs (FDPs)/conferences/workshops, such as ICCIS'23, ICDLAIR-2019, ATAL-FDP on "Cyber Security," FDP on "Software Defined Networks," and FDP on "Technical Writing: Tools and Techniques," to name a few. She also received funding from IEEE CIS for organizing Summer School in 2022 and 2023.